



Bosch, Núria. 2025. Categorical granularity in syntactic acquisition: A multilingual corpus study on the left periphery. *Glossa: a journal of general linguistics* 10(1). pp. 1–70. DOI: <https://doi.org/10.16995/glossa.23986>



Categorical granularity in syntactic acquisition: A multilingual corpus study on the left periphery

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The development of functional categories crosslinguistically is an extensively studied and debated topic (i.a., Radford 1990; Boser et al. 1992; Rizzi 1993; Guasti 1993; Clahsen et al. 1994; Friedmann et al. 2021; Heim & Wiltschko 2025). In this paper, I contribute to existing literature by exploring the explanatory potential in acquisition of a previously understudied formal notion – *categorical granularity* (following Biberauer & Roberts 2015; Song 2019; Biberauer 2019). I study the development of the left periphery with a multilingual corpus study on 10 children, across 5 languages. First, a novel contrast is observed: on the one hand, CP-structures emerge very early, favouring accounts with early development of CP. On the other hand, however, there is virtually no evidence in the production data to assume children are operating with a split, cartographic-type CP until a later stage. Second, I show that acquisition and structural height can mismatch: some structurally very high elements (topics, illocutionary complementisers) emerge very early in several children. I then argue that these empirical generalisations pose important challenges for several extant theories of syntactic development, especially those assuming (fully) innate categories. This is because contemporary maturational and continuity approaches posit universal categorical sequences of *fixed* (often cartographic) granularity. Some of them additionally predict a strong correlation between height in a cartographic tree and timing of acquisition (e.g., Friedmann et al. 2021). I interpret the preliminary results reported here to stress the analytical strengths of incorporating *changes in granularity* when modelling syntactic development. The data presented provides an initial impetus for a role of granularity in syntactic development and for a reconceptualisation of the latter in terms of emergent categories. I develop a neo-emergentist account of the patterns, probing Biberauer & Roberts’s (2015) emergent categorical hierarchy. I maintain that this is a productive way forward, and finish by considering how this neo-emergentist thinking could be extended in future work.



1 Introduction

An ongoing debate within generative discussions of syntactic acquisition concerns how and when functional categories (i.e., non-lexical categories encoding grammatical properties) are acquired.

A traditional distinction in the literature is between so-called *maturational* and *continuity* approaches; these contrast with respect to whether all/most knowledge encoded in Universal Grammar (UG) is available from the start, with the former positing *gradual* development of UG principles and hierarchies (see, e.g., de Villiers 2001, for a review). Among the many issues grappled with in these frameworks, two stand out as central:

- (1) Are functional categories innate or (partly) emergent?
- (2) If innate, to what extent is syntactic category formation directed by a biological program (maturation)?

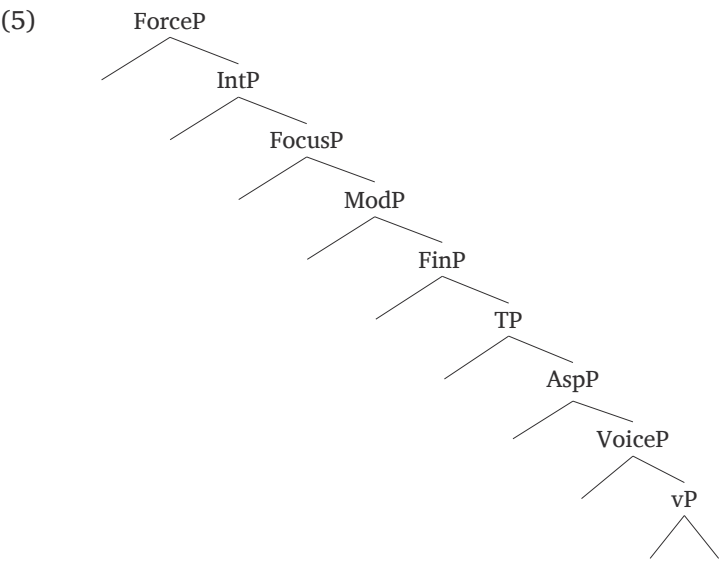
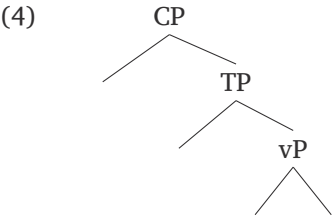
Given the wider commitments of the generative framework, (1) has thus far usually been answered by favouring innate categories (Diercks et al. 2023 is a recent exception). Nonetheless, the use of emergent categories in generative syntax and typology also has its proponents (e.g., Ramchand & Svenonius 2014; Ritter & Wiltschko 2014; Wiltschko 2014; Biberauer 2019; Larson 2021). With respect to (2), the choice between continuity and maturation remains highly debated; the former assumes complete access to UG content from the start, while the latter proposes instead gradual (maturation-guided) development of UG knowledge. Generative proposals work under a *tripartite* model of the structure of clauses where these are organised into three key clausal domains (e.g., Grohmann 2003) – *vP/VP*, which encodes thematic and argument-structural relations, *TP*, which establishes tense and/or other deictic properties, and *CP*, which encodes discourse information. Maturation and continuity proposals differ, then, in how many of these domains (if any) are available to learner at the start of grammar development.

Within maturational approaches, the dominant proposed *directionality* of development has arguably been bottom-up, where structurally lower elements mature first, e.g., *vP/VP* matures first, before *TP*, which in turn matures before *CP* (i.a., Radford 1990; Rizzi 1993; Friedmann et al. 2021). However, inward perspectives (where *CP* and/or *vP* mature first) have also been put forward, adducing empirical data that challenges bottom-up development (i.a., Galasso 2003; Tsimpli 2005; Heim & Wiltschko 2025). Therefore, different implementations of both continuity and maturation have been proposed, also reflecting the progressive addition of new theoretical machinery (e.g., cartography), to account for new developmental patterns. The wide range of approaches proposed to date (Strong Continuity, Weak Continuity, several articulations of maturation, etc.; see section 2) plausibly indicate the non-negligible, but still somehow *constrained*, variation in crosslinguistic developmental patterns and undoubtedly also the difficulty of modelling acquisition with this toolkit.

These contentious issues have remained to the present day. However, their prevalence has led to other, equally interesting empirical questions being addressed less. One of these questions forms the point of departure for this paper:

- (3) What is the *categorical granularity* of children’s representations at various developmental stages?

As for (3), the granularity (roughly, detailedness) of children’s syntactic knowledge in the literature thus far has typically been dictated by UG specifications. Work of cartographic orientation generally postulates a cartographic spine in UG, with the resulting acquisition theory being based on its projections (e.g., Westergaard 2009; De Lisser et al. 2017; Friedmann et al. 2021). On the other side, where acquisitionists posit more pared-down spines (e.g. those centred on a basic CP>TP>vP; i.a., Galasso 2003, Tsimpli 2005, Vainikka & Young-Scholten 2011), cartographic-type syntactic patterns are set aside in their acquisition studies. Two examples of more minimalist and cartographic-type functional projections are given in (4)–(5), respectively. This illustrates differences across frameworks in categorical granularity: cartographic-style approaches postulate richer functional structure that elaborates each of the three clausal domains. In both cases, children’s granularity is *fixed* across all stages, bound by the functional hierarchy in UG. As a result, (3) has generally not been treated as a direct object of study thus far.



But it does not follow that granularity should merely be taken for granted on *a priori* grounds. This is especially true if one acknowledges that there may be crosslinguistically variable *degrees of elaboration* of functional structure, like the left periphery, as underlined by several proposals (i.a., Ramchand & Svenonius 2014; Biberauer & Roberts 2015; Biberauer 2018; see section 2.4). Indeed, after positing a given fixed UG spine (e.g., a cartographic one), evidence in children's data is needed, across all stages, to *motivate* the postulation of, for example, an articulated CP-domain. Otherwise, a given granularity is simply presupposed prior to the relevant developmental study, and is thus not falsifiable unless explicitly interrogated. More specifically, we can ask, if we accept (some of) the arguments for fine-grained cartographic functional sequences: how and when do children acquire the rich functional hierarchies posited typologically (e.g., Rizzi 1997; Cinque 1999; Svenonius 2006) *and* when does evidence for this surface in their data? (3) is, then, a closely linked, but *separate*, question from (1) and (2).

Against this background, this paper has three goals:

Firstly, I propose that categorial granularity (henceforth, simply 'granularity') should not just be a theoretical prior, but must be *probed*, just like (1)–(2) are. The paper's primary goal, then, is to present an initial case study demonstrating the importance of considering (3). I summarise a study on the development of the left periphery in 10 children across 5 languages. A contrast emerges from the data: while CP-structures emerge from the earliest files, evidence for co-occurrence of CP-elements (motivating a Split CP-type structure) appears systematically at a later stage. This observation forms the basis of this paper.

The paper's second goal is to problematise the *directionality* component of (2), particularly once cartography is adopted: do crosslinguistic developmental patterns align with a directionality-driven approach to the acquisition of functional spines? I will argue that *multilingual* corpus data provides significant insights into the absolute pathways predicted by maturation, with the present data suggesting a negative answer to this question. Topicalisation and illocutionary complementisers provide the crucial empirical evidence to this effect.

The resulting empirical picture challenges, to different extents, most extant approaches to development, which either undergenerate or overgenerate. I suggest that *granularity* provides the flexibility needed to interpret the data.

The third and final goal is to provide an account of the data following *neo-emergentist* generative perspectives on development (following Biberauer 2011, *et seq.* and Biberauer & Roberts 2015),¹ as well as to highlight some predictive similarities and differences with other existing frameworks. This paper aims to show how, by conceiving of categorial development

¹ Neo-emergentist approaches adopt a maximally poor UG, modelling acquisition, to the greatest extent possible, with independently motivated general cognitive principles (so-called 'third factors' in Chomsky 2005). Crucially, however, their essence remains generative: systems are regulated by UG, Merge and formal features. These theoretical assumptions set the approaches apart from traditional emergentist approaches such as Construction Grammar

as (at least partly) emergent, we arrive at an approach to development where these otherwise problematic patterns fall into place. This is my final theoretical contribution.

The paper is organised as follows: section 2 summarises extant approaches to syntactic development. Here, I also identify the underexplored and testable aspect of syntactic development addressed above, namely a possible role for categorial *granularity*. I highlight the contrast with the focus of the (maturational) literature thus far – developmental *directionality*. Section 3 introduces the multilingual corpus study, which addresses the three important aspects of development outlined in (1)–(3); I summarise its methodology, diagnostics and results. This section demonstrates that current approaches (especially bottom-up maturational approaches; section 2.1) face important challenges: the data show very early emergence of several CP-structures, including structurally very high ones, but systematically late emergence of structures indicating the child is harnessing an elaborate, “split” CP. I take these patterns as initial evidence to motivate the adoption of granularity as a potentially productive tool when modelling development. I develop this proposal in section 4, where I also evaluate the approaches discussed in light of the data. Section 5 concludes.

2 Background: approaches to the acquisition of functional categories

Traditionally, approaches to syntactic development have been divided into two main kinds – maturational and continuity – depending on the amount of innate linguistic structure assumed to be accessible to the child at the start of the learning path. I begin first with maturational approaches.

2.1 Maturational approaches

“Incomplete structure” or *maturational* approaches argue that children’s grammatical knowledge is not (fully) available initially and that grammatical categories and principles appear gradually. The key proposal is to operationalise this “gradual” development in terms of syntactic maturation: a UG-given spine (whose exact articulation varies across proposals) and its categories develop (“mature”) with a specific *directionality*, determined by UG. The dominant directionality harnessed in maturational approaches has been a *bottom-up* one, whereby domains projected by structurally lower heads (e.g., ν P/VP) are predicted to be acquired before higher ones (e.g., CP). This hypothesis finds its roots in proposals such as Radford’s (1988; 1990) Small Clause Hypothesis and Rizzi’s (1993) Truncation Hypothesis.

Since the focus of this paper lies on the left periphery, I will concentrate on a more recent *cartographic* and left-periphery-centred interpretation of bottom-up maturation, namely

and usage-based theories in non-negligible ways, hence the prefix *neo*-emergentism (see Biberauer 2019: 59). Neo-emergentism is described in more detail in section 2.4.

Friedmann et al.'s (2021) *Growing Trees Hypothesis*. However, the results in section 3 will have implications not just for Growing Trees, but for any version of bottom-up maturation. In Growing Trees, the development of the left periphery is composed of two stages: at an initial Stage 1, the CP-domain is yet to develop. Only IP/TP and VP are available to the child, permitting, for instance, the production of inflection and A-movement. A later Stage 2 sees the development of the lower left periphery, up to QP (hosting, i.a., wh-elements). The entire cartographic hierarchy, from TopP until ForceP, becomes available at Stage 3, when, e.g., topicalisations and embedding are expected to appear.² The full developmental trajectory proposed is schematised in **Figure 1**. This thereby predicts intermediately late emergence of wh-questions (before higher projections, but *after* VP/IP) and very *late* emergence of elements like topics or complementisers.

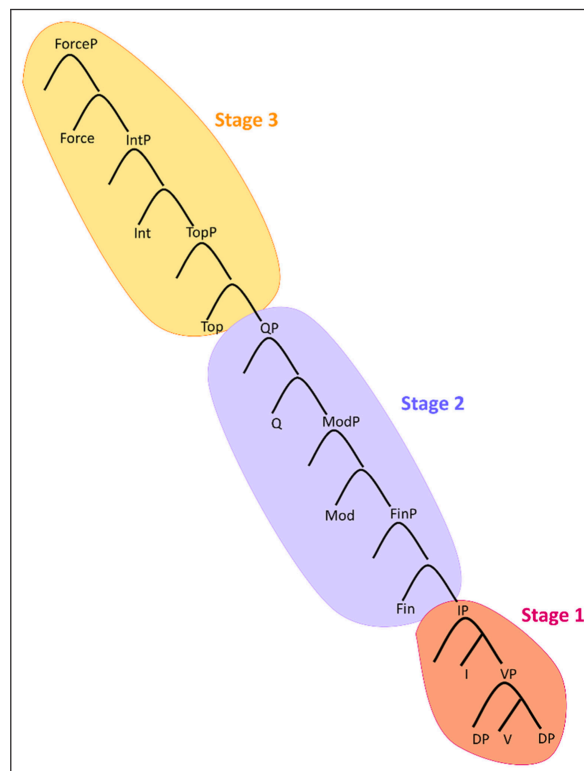


Figure 1: Stages in the Growing Trees Hypothesis (Friedmann et al. 2021: 12).

² Unlike Radford (1988; 1990), Stage 1 already contains functional categories (IP/TP). Friedmann et al. (2021: 27) consider the possibility of an earlier phase with a bare VP lacking IP, ultimately concluding there is no evidence for it in their data as inflection is present (age range 1;6–6;1). They do not consider the possibility that the attested inflection does not reflect adult-like representations. Note, too, that in the full clausal spine in **Figure 1** only the left periphery is cartographic. How the child would acquire more elaborate versions of the Tense-Aspect-Mood domain (see also Cinque 1999, cf. too De Lisser et al. 2017), or of the vP-domain and event-structure (e.g., Ramchand 2008) remains an open question.

The above has discussed *bottom-up* maturation. Now I turn to *inward* maturation,³ which has received much less attention than the dominant bottom-up approaches. Simplifying grossly, inward approaches can be defined as approaches that assume development to begin at *vP*/*VP* and *CP* (the phasal edges, from the perspective of the *CP* > *TP* > *vP* functional sequence), before maturation of *TP*. Under inward maturation, then, we expect early development of (some) *CP* knowledge (alongside (some) *vP* knowledge).

Arguably, the first maturational approach that, in retrospect, argues for this progression can be traced back to Galasso's (2003) "Empty Middle" approach for child English, which proposes a revision of Radford's (1990) Small Clause hypothesis: first, *CP* > *VP* are acquired; then, the full *CP* > *IP* > *VP* matures. The approach assumes a *VP* structure for basic 'telegraphic' declaratives (with Radford), *CP* > *VP* for basic interrogatives at his stage 1 and, finally, the full blown *CP* > *IP* > *VP* at stage 2, when inflection, auxiliaries and inversion start being acquired. For a slightly similar hypothesis, see Tsimplici (2005), whose maturational pathway hinges instead on the development of uninterpretable vs. interpretable features.

More recently, Heim & Wiltschko (2025) outline an explicit inward-growing maturational approach, the *Inward Growing Spine Hypothesis*, adducing support from the development of the English discourse particle *huh*. This hypothesis draws on two associated models of clausal organisation: the Universal Spine Hypothesis (Wiltschko 2014) and the Interactional Spine Hypothesis (Wiltschko 2021). These are jointly represented in **Figure 2**.

In this approach, the thematic and categorisation-based domain (centring on Wiltschko's 2014 *Classification* head) and the interactional and speech-act-based domain (Wiltschko's 2021 *Responding* head) are hypothesised to develop first, along with *Linking* (broadly speaking, the *CP*, hosting, e.g., *wh*-words and complementisers). The subsequent steps involve gradual 'unfolding' into an adult-like spine: first, the child develops the *Linking* layer further into two projections (a *Grounding* layer, above *Linking* and an *Anchoring* layer, below *Linking* and roughly equivalent to *TP*). Finally, the *Anchoring* layer is divided into a deictic and a perspectival anchor (e.g., *Tense* and *Aspect*); *Grounding* is divided into a speaker-oriented (*Ground_{Spkr}*) and an addressee-oriented layer (*Ground_{Addr}*) (Heim & Wiltschko 2025: 15ff).

This is visualised in **Figure 3**. Since the present paper largely does not discuss interactional language concerning most of the speech-act layers assumed by Heim & Wiltschko, I will not be able to test this specific hypothesis, but a more general version of an inward-growing approach (e.g., Galasso 2003), discussed in section 2.5. However, I will briefly return to Heim & Wiltschko (2025) in section 4 in light of the results observed.

³ Echoing Heim & Wiltschko's (2025) *Inward Growing Spine Hypothesis*.

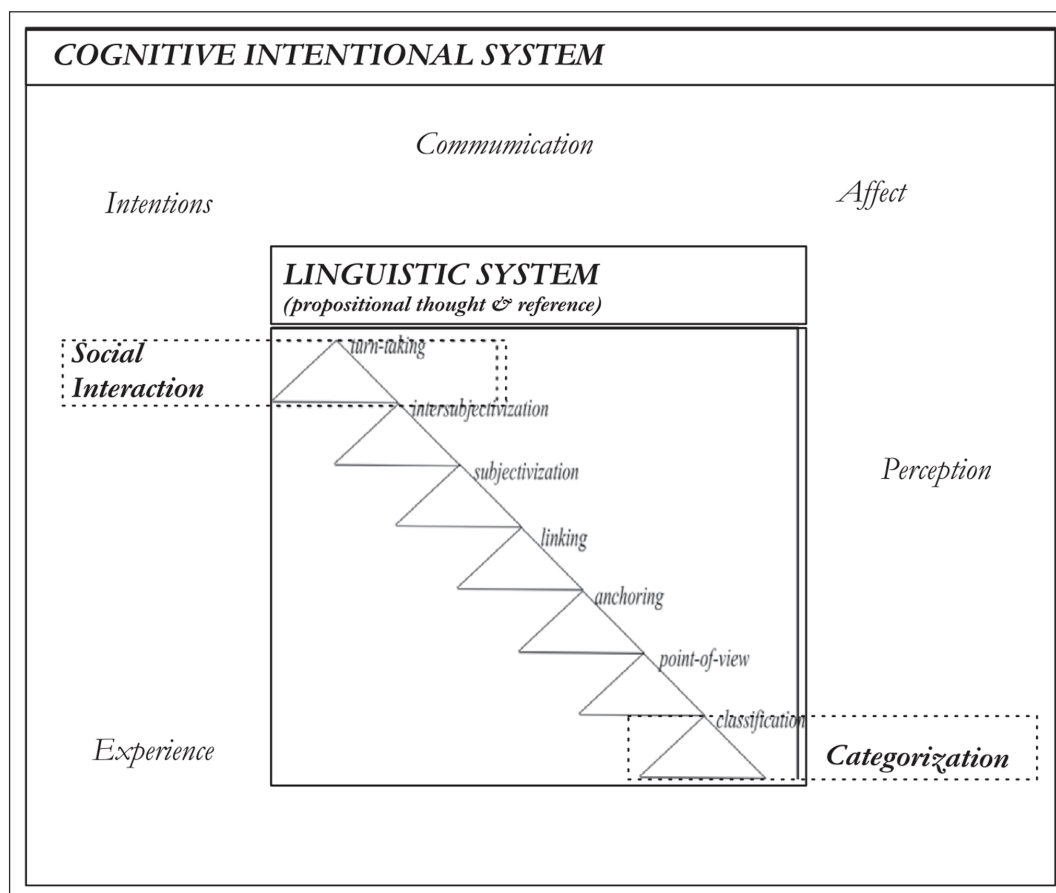


Figure 2: The Bridge Model (Hinzen & Wiltchko 2023: 75).

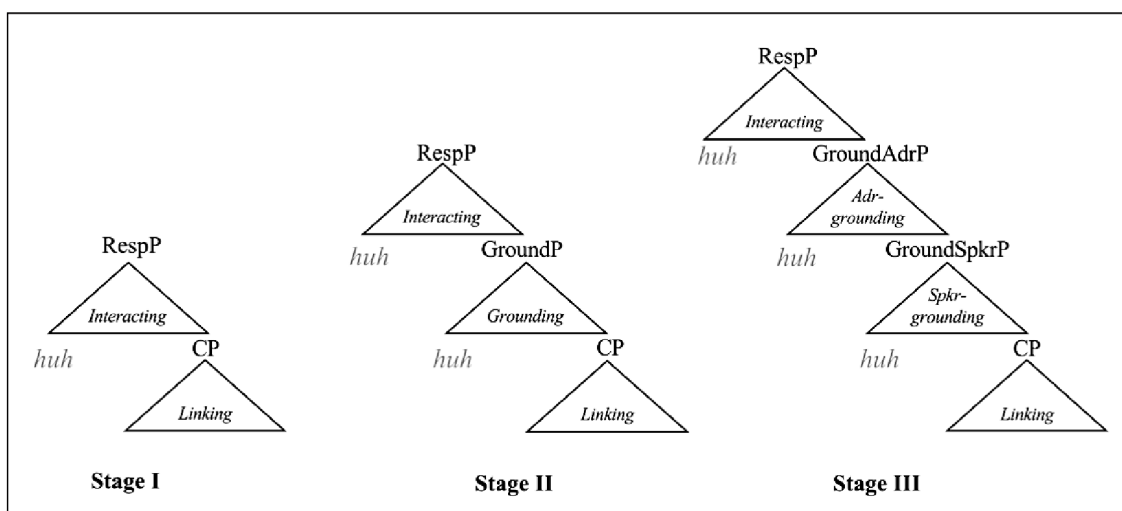


Figure 3: Stages of the Inward Growing Spine Hypothesis, applied to *huh* (Heim & Wiltchko 2025: 23).

2.2 Continuity approaches

Continuity approaches, on the other hand, claim that the functional structure of children's initial grammar is not significantly different from adults' grammars. The extent to which child and adult grammars are taken to be isomorphic varies, with both so-called Weak Continuity (i.a., Clahsen 1990; Clahsen et al. 1994; 1996) and Strong Continuity approaches (i.a., Pinker 1984; Boser et al. 1992; Hyams 1992; 1994; Verrips & Weissenborn 1992; Poeppel & Wexler 1993) having been put forward. For example, Poeppel & Wexler (1993) and Boser et al. (1992), both on the basis of V2 data in child German, suggest that the child grammar includes the categories and operations in the adult grammar (a CP domain which hosts V2-topics in its specifier, and moved finite verb in C; den Besten 1983) from the start, with no period in the production displaying a target-deviant stage with finite verbs in OV position.⁴

A relevant cartographic continuity perspective on the left periphery is Westergaard's (2009) Micro-cues approach. Westergaard (2009) argues for sensitivity to fine-grained cartographic and information-structural content from the earliest stages of acquisition; in particular, children seek "micro-cues" in the Primary Linguistic Data (i.e., the input) that signal the specific configuration of the adult grammar they are exposed to (e.g., the micro-cue for Norwegian (non-V2) embedded *wh*-questions is hypothesised to be [_{whP} *wh* ... [_{VP} V]], ch. 3). The claim, then, is that children are successful in deploying the knowledge distilled from these micro-cues from the earliest multi-word combinations.

As a result, all continuity approaches share the intuition that functional structure is an early phenomenon, with all of CP, TP and *v*P/VP being accessible (in some form or another) from the start. More specifically for present purposes, they predict early attestation of CP-structures.

2.3 Interim summary: a prediction with innate categories

Taking stock of the past two sections (maturation and continuity), maturation postulates *universal* orders of acquisition of structures: UG not only specifies the functional spine shaping all languages, but also the specific order in which its categories will be acquired. Notice, crucially, that what changes throughout development is *not* the UG-given tree; it is how much of that UG-given spine children have access to at different stages, subject to a given maturational directionality (bottom-up or inwardly). The *granularity* of the syntactic categories and the spine postulated is *fixed* or *static*: if a cartographic spine is the UG prior, then children will acquire cartographic domains (internally layered with multiple projections) once they mature. *Mutatis mutandis*, the same point holds for the continuity approaches just discussed, which assume full access to UG

⁴ Note that continuity approaches do not suggest that there is no language-specific grammatical development with respect to functional structure; instead, the universal functional spine on which an L1 grammar is built is accessible from the outset. L1-specific properties may be hypothesised to be acquired, for example, via so-called "parameter setting".

content from the start; Westergaard (2009: 104), for example, argues that children harness cartographic left-peripheral knowledge “from the onset of multiword utterances”. As above, children operate, constantly, with the granularity specified by the analyst as being provided by UG (here: cartographic).

The one exception here is Heim & Wiltschko (2025), who already adopt a more explicit notion of granularity, via successive divisions into their full, adult-like speech-act projections; however, they explicitly do *not* adopt a cartographic CP. This is something I return to in the next section and section 4.

Importantly, as also noted in passing by Westergaard (2009), Moscati & Rizzi (2021: 16) and Moscati (2023: 3), this theoretical commitment makes a *testable prediction* with respect to children’s categorial granularity: once (or soon after) CP develops, we should observe (some) evidence that the child is indeed harnessing a CP with elaborate structure. Our expectation, therefore, is that we should see evidence in the child data, once they start producing/comprehending CP-structures, that postulating a fine-grained CP-domain is *justified*, and is not merely taken for granted on theory-internal grounds. This prediction will be a key focus of the corpus study in section 3.

2.4 Neo-emergentist approaches

The approaches introduced thus far generally adopt what we could term a “fixed granularity” perspective on acquisition, with children’s categorial granularity being specified by UG. This assumption that categorial granularity *never* changes throughout development is, however, not a necessary one. This section discusses so-called *neo-emergentist* approaches (Biberauer 2011, *et seq.* and Biberauer & Roberts 2015), one of whose central claims is that *increasing* (i.e., *flexible*) *granularity* is, in fact, a key aspect of syntactic development.

Capitalising on the consequences and opportunities that a minimal UG brings, Biberauer (2019) endorses a Three Factors approach, in turn proposing an explicit third-factor principle guiding acquisition (Maximise Minimal Means, MMM). Importantly for present purposes, she introduces the essence of a *neo-emergentist* approach: acquirers are steered by a minimal UG, containing Merge, Agree and some notion of $[F]$, and by MMM. The latter is a general-cognitive bias leading learners to make maximal use of minimal resources. This leads acquirers to generalise already-acquired knowledge as much as possible and only postulate new formal content (e.g., $[F]$ s) if required by the Primary Linguistic Data and if it fulfils a *contrastive* role in the system (for greater detail on this approach and MMM, see Biberauer 2019). The essence crucially remains fundamentally generative: systems are “guided” by a minimal UG, making systems both Merge- and formal-feature-regulated (Biberauer 2019: 59). The emphasis, however, lies on exploring the empirical productivity of impoverishing UG, in line with Chomsky (2005). In contrast

to maturational approaches, neo-emergentism suggests that development is best modelled via *emergent* syntactic categories.

Specifically, I will focus on Biberauer & Roberts' (2015) *emergent categorial hierarchy*. Firstly, according to Biberauer & Roberts (2015), crosslinguistically, clauses can be analysed at different levels of 'magnification' or 'granularity'. Having briefly discussed granularity in broad conceptual terms thus far, I define it more explicitly below, following Song (2019: 21):

(Syntagmatic) granularity: A sequence of combinatorially oriented categories may be collapsed into a single category (e.g., $X - Y - Z \rightarrow A$), and a single category may be split into a sequence of categories (e.g., $X \rightarrow X_1 - X_2 - \dots - X_n$).

For example, the relation between the Core Functional Category C (Chomsky 2000; 2001) and the cartographic sequence Force–Topic–Focus–Fin (Rizzi 1997) is one of (syntagmatic) granularity; the former can be split into the latter and, vice versa, the latter can be collapsed into the former. The major granularity levels I assume throughout this paper are the following (Biberauer & Roberts 2015: 4), from less to more fine-grained (as the nominal domain is orthogonal to the present study, I restrict exposition to the clausal domain):

- (6) **Extended Projection** (V) > **phase** (C, v) > **Core Functional Category** or **CFC** (C, T, v) > **“cartographic field”** (e.g. Tense, Mood, Aspect, Topic, Focus) > **semantically distinct head** (as in Cinque 1999; Speas & Tenny 2003; Frascarelli & Hinterhölzl 2007; Haegeman & Hill 2013, among others).

At the highest level of organisation, we find the Extended Projection (EP) in the sense of Grimshaw (1991: *et seq.*), referring to a maximal projection which projects from a category that it shares categorial features with. Thus, the whole clausal EP is C–T–V, with all categories sharing the features [$+V$, $-N$]. Here, the only featural contrast in the system may be between predicates and nouns ($\pm V$ or $\pm N$). At a second granularity level, we encounter Phasal distinctions (C, v; Chomsky 2008). These are located at a coarser-grained level than the three Core Functional Categories C, T and v (Chomsky 2000; 2001), given that T is not inherently a phase. Then we encounter the finer-grained cartographic fields (e.g., Force–Topic–Focus–Fin) and semantically more fine-grained heads such as those in Cinque (1999). The former represent iterations of C-positions (i.e., sequences of heads with [C]), each associated with a grammaticalised semantic feature (e.g., Topic with [TOPIC]). The exact elaboration of the left periphery is therefore taken to be a matter of crosslinguistic variation, just like in earlier and contemporary work (i.a., Giorgi & Pianesi 1997; Bobaljik & Thráinsson 1998; Walkden 2017; Hsu 2017; Larson 2021). I will focus in particular on the contrast between C as a CFC/Phasal category vs. its potential cartographic elaborations. In parametric terms, the cartographic level of granularity corresponds to the microparametric level of variation, while coarser-grained levels recruit macro/mesoparametric options.

In turn, (6) can be reformulated as an emergent categorial hierarchy, which unifies and identifies overall commonalities between independently proposed hierarchies and categorial sequences in the literature. The hierarchy thus begins at an initially ‘undivided’ categorial space and gradually increments the level of grain with which the system is analysed (Biberauer & Roberts 2015: 6):

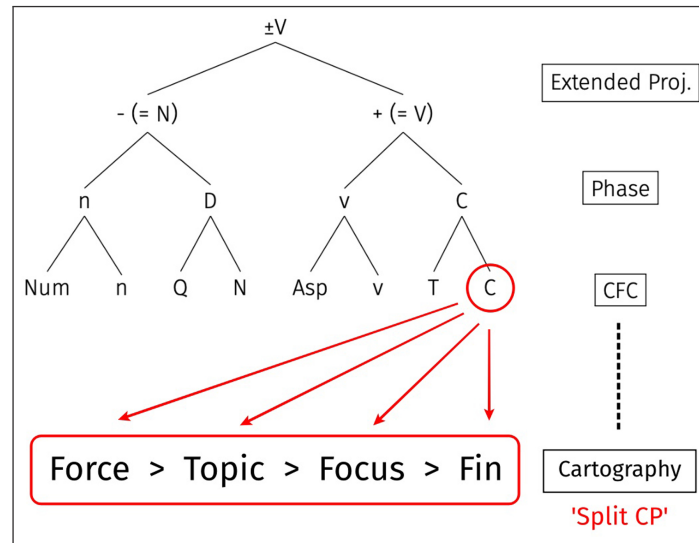


Figure 4: Biberauer & Roberts’s (2015) emergent categorial hierarchy, adapted to include a cartographic CP.

The key for present purposes is that **Figure 4** can be conceptualised as a learning path. The emergent hierarchy is regulated in a contrastivist manner (i.a., Hall 2007; Dresher 2009; Cowper & Hall 2014), with acquirers postulating ever finer-grained categorial distinctions once contrastive features are detected in the input. Although not represented in the original hierarchy in Biberauer & Roberts (2015: 6) for simplicity, it is not until a later stage, after the emergence of the CFCs, that the approach expects cartographic heads and semantically distinct heads (as in Rizzi 1997, Cinque 1999, respectively) to emerge and further subdivide each of the CFCs. Being a phasal domain, this approach predicts CP to develop *before* TP, like inward approaches, but proposes this CP will remain “basic” (non-split) in organisation until a later stage. I will centre on the prediction that acquirers will first entertain a “basic” CP-domain (of the Phasal or CFC kind) at an early stage, before cartographic-type knowledge is considered (space considerations preclude discussion of TP, but for discussion see Bosch 2023a, where the emergence of TP-structures is analysed in light of this hypothesis). Put differently, acquisition is such that microparametric distinctions are not posited unless a macro/mesoparametric analysis of the Primary Linguistic Data cannot be maintained (Biberauer & Roberts 2015: 14). The acquirer is *conservative* when it

comes to positing higher levels of granularity and formal features, but *liberal* when generalising already-existing formal distinctions (viz. Feature Economy, Roberts & Roussou 2003, and Input Generalisation, Roberts 2007/2021; and Maximise Minimal Means, which unifies the two).⁵

Biberauer & Roberts (2015) converge, then, with the already-discussed inward-growing and continuity approaches (section 2.1) which have also, in various forms, argued for early emergence of the CP-domain, and against a strictly bottom-up approach to development. However, and in contrast to the approaches discussed thus far, evidence for cartographic-type knowledge should come visibly late in the production data – this is because categories *granularise* in development, elaborating on developmentally-prior knowledge. Therefore, this perspective explicitly envisages *changes* in granularity during development, which correlate with the acquisition of macro/microparametric knowledge. Crucially, some previous work already offered very suggestive speculations in this direction. Roeper & de Villiers (2011: 9) noted that “one possibility [...] is that the child begins with a proto-CP, and it undergoes an actual process of Splitting as suggested by Hollebrandse & Roeper (1998) to end up with a distinct series of nodes with different functions”. A similar logic is adopted in Soares (2006) for European Portuguese data.

The current approach is also in principle synergetic with Heim & Wiltschko (2025), which, as noted above, incorporates some notion of granularity, e.g., their final stage involves “subdividing” their Ground head into Ground_{Addr} and Ground_{Spkr}. Given that their hypothesis is not cartographic and largely concerns projections above CP, I will mostly set it aside. However, future work should consider the clear overlap between the approach proposed here, Wiltschko’s (2014; 2021) categorical template and Heim & Wiltschko’s (2025) Inward Growing Spine.

Overall, then, the intuition we will pursue in the corpus study is as follows: granting that various degrees of granularity are needed to capture crosslinguistic variation (per Biberauer & Roberts 2015), we will not, *a priori*, assume a specific granularity (in UG or elsewhere) as the starting point for children. Instead, we “let the data decide”, taking child productions to be informative as to the granularity acquirers are most likely to be harnessing at various stages. This is important, insofar as it inquires into not just the “earliness” of functional domains (e.g., CP, TP, etc.), but *also* their granularity throughout development. The latter represents an almost

⁵ This approach therefore assumes parts of the input do *not* qualify as intake at certain points in development (e.g., structures requiring a fully-fledged CP), for example because their featural complexity renders them inaccessible early on (see Gagliardi 2012; Lidz & Gagliardi 2015). The key is that these ‘Split CP-structures’ become acquisitionally possible (and so can be ‘triggered’ by the input) once children have first identified the need for a generic C-domain in their language. This still begs the question of how Split CP-structures would be expected to be syntactically comprehended before their full representation is acquired (if comprehended at all), as an anonymous reviewer points out. The same question arises for other approaches assuming “incomplete” representations during early development (section 2.1). It is well beyond of the scope of this paper to address this, but at the very least, this is also an empirical question, which could be explored once experimental studies probing these theoretical approaches are carried out (see section 4.2).

undiscussed aspect of syntactic representations in development and a novel way of approaching corpus data.

2.5 Predictions

Summarising the preceding discussion, the architectural differences of the approaches introduced, as well as their predictions for the left periphery, can be broken down in terms of the following four criteria:

- (i) Biological locus of syntactic categories (innate vs. emergent).
- (ii) Postulation of syntactic maturation and its directionality (if applicable).
- (iii) Acquisition of CP (early vs. late).
- (iv) Fixed vs. flexible granularity.

These are compared in **Table 1**, which incorporates the approaches introduced thus far.

Approach	Maturation	Locus	Acquisition of CP	Granularity
Bottom-up approaches (Growing Trees)	Bottom-up	Innate	Late (two-stage)	Fixed
Inward approaches	Inwardly	Innate	Early	Fixed
Continuity	✗	Innate	Early	Fixed
Neo-emergentism	✗	Emergent	Early	Flexible

Table 1: Architectural differences between approaches to syntactic development.

All four groups of approaches make clearly testable predictions. Inward-growing, continuity and neo-emergentist approaches anticipate early acquisition of (some) CP-structures, irrespective of their structural height in the left periphery. Bottom-up approaches, on the other hand, maintain that CP is a late(r) development, relative to TP. More specifically, the hypothesis under consideration here (Growing Trees) suggests this development is two-stage, such that the structurally highest heads (Topic, Force, etc.) mature after left-peripherally lower elements (like QP/FocP, which hosts, among others, *wh*-questions). I focus in particular on the commitment that there will be a relative order of emergence of left-peripheral structures correlating with structural height. Importantly, neo-emergentism is the only approach making explicit predictions with respect to granularity. In this approach, granularity *increases* throughout development, by virtue of its emergent categories, and this should accordingly be reflected in children's competence and production: as such, CP-structures (on their own) may emerge early, but we do *not* anticipate more specialised (granular) CP-structures to be combined until a later stage. The relevant split CP-structure is assumed to be featurally more complex, so acquisitionally harder. Notice therefore

that neo-emergentism does *not* adopt *syntactic* maturation,⁶ in the sense that the relevant finer-grained categories are hypothesised to be emergent, not innate. All other approaches typically do not entertain changes in granularity; the granularity children operate with is fixed (hard-wired) by the functional structure in UG. This distinction between predicting “fixed” vs. “flexible” granularity is operationalised in the diagnostics introduced in section 3.2.

3 The corpus study

3.1 Children studied

The spontaneous speech of 10 children across 5 languages in the CHILDES database MacWhinney (2000) was analysed; this involved 2 children per language, which included Catalan, Italian, Spanish, German and Dutch. Overall, the period in which children’s production was analysed ranged from 0;11 to 5;02, across all children.

The details of the children studied, including their word-based MLU (MLUw), are reported in **Table 2**. Henceforth, all examples and tables will be presented in the order of languages and children given below.

Language	Corpus	Child	Files analysed	Age range	MLUw range	Total words	Total utterances
Catalan	Serra-Solé	Laura	19	1;07–4;00	1.03–3.47	16651	7576
		Gisela	21	1;07–4;02	1.02–3.51	11132	4753
Italian	Calambrone	Martina	13	1;07–2;07	1.26–2.69	7569	3586
		Rosa	21	1;07–3;03	1.27–3.24	14659	6614
Spanish	Llinàs-Ojea	Irene	59	0;11–3;02	1.0–5.13	38234	11908
	Montes	Koki	13	1;07–2;11	1.96–3.61	11773	4276
German	Miller	Kerstin	37	1;03–3;04	1.09–2.89	28609	16686
		Simone	50 ⁷	1;09–2;09	1.52–4.89	45407	21371
Dutch	Groningen	Josse	28	2;0–3;04	1.2–4.01	30066	11120
	van Kampen	Sarah	50	1;06–5;02	1.07–6.07	49465	17408

Table 2: Children studied in the CHILDES database and summary information (Serra & Solé 1989; Cipriani et al. 1989; Llinàs-Grau & Ojea 2000; Montes 1987; Miller 1979; Bol 1995; van Kampen 2009).

⁶ This assumption of course does not entail a rejection of *neurobiological* maturation; the above concerns only *syntactic* maturation, understood as pre-wired development of (some of) the domain-specific repertoire in UG. The approaches presented – both continuity and neo-emergentism – are compatible with approaches that try to account for patterns of syntactic development in terms of maturation of neurobiological substrates (e.g., see Snyder et al. 2024). The key distinction here is between what Lust (1999: 122–123) termed the *General* Maturation Hypothesis and the *Specific Grammatical* Maturation Hypothesis. This paper focuses on the latter; the former is not contentious.

⁷ Being the densest corpus of all, only the first 50 files in Simone corpus were analysed, out of the 73 recordings in total. All diagnostics studied are produced by Simone over these 50 files. For the remaining children, the entire corpora were studied.

I will adopt MLUw as the guiding developmental metric throughout, over age, given the high variability observed among children of the same age with respect to their linguistic development (see, for example, the discussion in Paradis & Genesee 1997; Caprin & Guasti 2009). This will also facilitate crosslinguistic and inter-speaker comparisons, as well as the replicability of these findings. The choice of word-based MLUs over morpheme-based metrics is due to two considerations: (i) not all corpora in CHILDES are morphosyntactically tagged, meaning morpheme-based MLUs are not available in practice; (ii) morpheme-based MLUs are more prone to annotation errors and subjectivity/arbitrariness, as the annotator has to establish which morphemes are productive at various stages (Ezeizabarrena & García Fernández 2017). Nonetheless, age will still be reported. Some notes of caution about the stability of MLU as a metric (either word- or morpheme-based) will be offered in section 4.

3.2 Research questions and diagnostics used

I conducted a multilingual corpus study to probe the following empirical and theoretical questions, synthesised from the preceding theoretical review in section 2:

- i. How early do CP-structures emerge in the child production data? Are these timings in accord with the theoretical approaches introduced (continuity, inward maturation, bottom-up maturation, neo-emergentism)?
- ii. Do left-peripherally high structures emerge later than lower CP-structures, as in Friedmann et al. (2021)?
- iii. Do children display knowledge of cartographic-type distinctions within CP from their earliest uses of CP-material? If not, does the data follow the ‘increasing granularity’ prediction by neo-emergentist approaches, where evidence for Split CP-structure should emerge comparatively late in production?

As emphasized in sections 1 and 2, *justifying* the granularity of the representations assumed for children at various stages is arguably as crucial as determining which domains (whether cartographic or not) are acquired first. This is the novel research question I seek out to address in the next sections. I do so by introducing a new methodological distinction in the data collection between ‘CP-structures’ vs. ‘Split CP-structures’, described below. Research question (iii) is clearly ambitious, I do not aim to provide a conclusive answer to (iii) in this paper, but rather to introduce relevant empirical data that serves as a springboard for future work on this topic.

3.2.1 CP-diagnostics

The study contrasted two sets of structural diagnostics for children’s knowledge of the left-periphery. First, I searched for structures standardly taken to involve the CP-domain. These included V-to-C (V2), wh-questions, yes/no questions, topicalisation and focalisation, illocutionary complementisers and embedding markers.

Evidence for V-to-C movement (or a distributional distinction between finite/non-finite verbs in V2/V-final position) in Dutch and German children will be taken to indicate an emerging CP-domain. More precisely, the study investigates evidence in the same recording of both non-finite verbs in a non-raised (or base) position with OV-order as well as of finite verbs which can be shown to have raised outside of the verbal domain. If the initial constituent is the subject or some non-object topic, these display surface VO-order when the verb is finite. This is illustrated below, where (7a), (8a), (8b) exemplify verb-raising of *haben/lust/eet*, shown by the absence of surface OV-order, and (7b) displays a combination of a finite modal in its V2-position (*willen*) and a non-finite verb *trinken* following its object *Kola*:⁸

(7) *German*

- a. Sie haben Glück
they have.3PL luck
'They are lucky.'
- b. Ich will Kola trinken
I want.1SG Cola drink.INF
'I want to drink coke.'

(8) *Dutch*

- a. Ik lust geen kaas
I desire.1SG no cheese
'I don't like cheese.'
- b. Nu eet ik een boterham
now eat.1SG I a sandwich
'Now I eat a sandwich.'

(7a)–(8a) instantiate cases of subject-initial structures with finite verbs. These structures have been analysed as involving raising of the subject and verb to the T-domain only (Zwart 1993; 1997; *et seq.*). I will, nonetheless, include subject-initial declaratives as evidence of knowledge of V2 (following, i.a., den Besten 1983; Schwartz & Vikner 1989; 1996; van Craenenbroeck & Haegeman 2007), on the grounds that exclusion of subject-initial sentences would effectively reduce the V2-diagnostic to the (non-default) topicalisation diagnostic. This is undesirable, as it would not control for a scenario where knowledge of the V2-system emerges *before* the first productions of topicalised non-subject elements. This point also becomes important in light of literature noting an initial stage in the acquisition of V-to-C in Germanic where children omit the first-position element, yielding surface V1 (Evers & van Kampen 2001; van Kampen 2010).

⁸ Unless otherwise noted, the examples provided are mine. For Italian, German and Dutch, they were also checked with native speakers.

Below I illustrate examples of wh-questions (with a wh-element raised to SpecCP, with subject-verb inversion):

- (9) *Catalan*
Quan comença el partit?
 when begin.3SG the match
 ‘When does the match start?’
- (10) *Italian*
Perché l’hai cambiato?
 why CL.DO = AUX.HAVE.2SG change.PTCP
 ‘Why did you change it?’
- (11) *Spanish*
Como cantó María?
 how sing.PRFV.3SG Mary
 ‘How did Mary sing?’
- (12) *German*
Wo hast du gegessen?
 where AUX.HAVE.2SG you eat.PTCP
 ‘Where have you eaten?’
- (13) *Dutch*
Wat doen we vanavond?
 what do.1PL we tonight
 ‘What are we doing this evening?’

Although analysed and quantified, wh-less questions of the sort in (14), attested in child Germanic languages, are not considered as sufficient evidence for emergence of the CP-domain. I required other CP-diagnostics to have *also* emerged for us to conclude that the CP-domain might be in place (this decision only became relevant in one child, Sarah).⁹ Questions with a wh-word but no verb were excluded.

- (14) *German* (Schmerse et al. 2013: 658)
 Macht das Pferd?
 do.3SG the horse
 ‘What is the horse doing?’

⁹ This requirement may be unduly conservative. Santelmann (2003) argues for wh-less questions in early Swedish that they display evidence of CP-level structure (see also Evers & van Kampen 2001; van Kampen 2009, on Dutch).

In Germanic languages, yes/no questions were included, given that these involve verb-raising to the C-domain. In contrast, polar interrogatives in Romance are typically marked via intonation only, with inversion having been lost for most modern Romance varieties (Giurgea & Remberger 2016). Note that VS-order as a marker of polar interrogatives has been retained in Spanish, coexisting with intonation-only marking (e.g., *¿Está María en casa?* vs. *¿María está en casa?* ‘Is Mary at home?’; Giurgea & Remberger 2016: 865). However, this order is syntactically ambiguous between a structure with raising to C (inversion properly understood) and one where the verb does not move from T and the subject remains in a lower position. For this reason, too, I exclude all polar interrogatives in Romance.

Topicalisations and focalisations are exemplified in (15)–(19). In Romance, these also include Clitic Left-Dislocation (CLLD), where the topic is accompanied by a clitic (15)–(16), and other cases of contrastive or informational foci (Laka 1990; Rizzi 1997), where a clitic is not required (17).

(15) *Italian*, CLLD

Questa la compro io
 this CL.DO = buy.1SG I
 ‘This one I’m buying.’

(16) *Catalan*, CLLD

I **a mi** em donaràs un regal?
 and to me CL.IO = give.FUT.2SG a present
 ‘And will you give ME a present?’

(17) *Spanish*, Contrastive focus

Arroz he comido (no pasta)
 rice AUX.HAVE.1SG eat.PTCP not pasta
 ‘I have eaten RICE (not pasta).’

(18) *German*, Topicalisation

Das glaube ich nicht
 that believe.1SG I not
 ‘That, I don’t believe.’

(19) *Dutch*, Topicalisation

Nu heb ik mijn bal vergeten
 now AUX.HAVE.1SG I my ball forget.PTCP
 ‘Now I have forgotten my ball.’

Knowledge of embedding was diagnosed via finite embedding structures containing overt embedding markers (e.g., complement clauses, relative clauses, conditionals, temporal, causal, purpose clauses, etc.). Non-finite embedded clauses were not counted as evidence for CP. Embedding markers are abbreviated in the tables in section 3.4.1 as ‘Embed’.

Finally, a further type of main-clause complementisers tracked in the Romance languages are so-called *illocutionary* complementisers, present primarily in Ibero-Romance (see Corr 2016). Illocutionary complementisers involve instances of main-clause complementisers (such as Cat./Sp. *que*) that do not function as subordinators heading a complement clause, and instead surface in main-clause contexts with a range of illocutionary functions (Corr 2016; 2022). Adopting Corr’s (2016) terminology, the item *que* in (20a) generates an exclamation (EXCLAMATIVE *que*); a quotation in (20b) (QUOTATIVE *que*); and in (21a), it contextualises the preceding utterance information, resulting in a conversational move (CONJUNCTIVE *que*). In Standard Eastern Ibero-Romance, matrix polar interrogatives can also be introduced with *que*, as in (21b) (INTERROGATIVE *que*). This means interrogative *que* is sanctioned in Catalan, but is rarer in Spanish.

(20) *Spanish*

- a. ¡**Que** casi sube y rompe la estantería!
 that.EXCL nearly climb.3SG and break.3SG the shelves
 ‘He/she nearly climbed and broke the shelves!’
- b. **Que** no lo pienso hacer
 that.QUOT not CL.DO = think.1SG do.INF
 ‘(I’ve said) I’m not going to do it.’

(21) *Catalan*

- a. Assenta’t **que** et pentinaré
 sit.down.IMP.2SG = CL.REFL that.CONJ CL.IO = comb.FUT.1SG
 ‘Sit down, I’ll brush/comb you.’
- b. **Que** em pots ajudar?
 that.INT CL.IO = can.2SG help.INF
 ‘Can you help me?’

Although most robustly attested in Ibero-Romance, some main-clause complementisers are also attested in Italian (e.g., *Dammene un altro **che** lo mettiamo qui* ‘Give me another one, we’ll put it here.’, ***Che** ciascuno scelga una carta!* ‘Everyone choose a card!’; Giurgea & Remberger 2016: 874). Therefore, they will also be included in the data analysis. Illocutionary complementisers are typically analysed as being located very high in the left periphery; in a cartographic approach, in Force (like embedded-clause complementisers; Rizzi 1997), and, very frequently, in higher

Speech Act heads, above Force (see, e.g., Corr 2016; 2022; Cruschina & Remberger 2018; Kocher 2022). These complementisers are abbreviated in the tables as “Illoc”.

3.2.2 Split CP-diagnostics

The diagnostics in the previous section would thus help establish children’s command of CP. Note, crucially, that they do not *per se* provide direct insights into the granularity of children’s representations, bar via theory-internal assumptions. They simply provide indications for CP knowledge and, as such, any results obtained could be compatible with cartographic and more minimalist frameworks. Therefore, a second set of novel diagnostics was used to probe specifically syntactic granularity. As Moscati & Rizzi (2021) note, the most direct evidence for knowledge of a fine-grained CP-domain would involve cases of co-occurrence of left-peripheral elements. This is the only diagnostic type adopted here for these structures and I take them to provide evidence for a CP-system that is additionally populated by a set of separate projections. I will refer to these diagnostics as “Split CP-diagnostics”.

In Romance, these will most frequently involve instances of topics preceding *wh*-phrases (22) or cases of recursive topics (24)–(25). A possible structural representation of (22) is given in (23).

(22) *Catalan*

La Júlia, on ha anat?
the Júlia where AUX.HAVE.3SG go.PTCP
‘Júlia, where has she gone?’

(23) [_{TopP} La Júlia [_{Q/FocP} on ha [_{TP} ... [_{VP} anat]]]]

(24) *Italian*

Questo, a te, ti spaventa
this to you CL.IO = scare.3SG
‘This, it scares YOU.’

(25) *Spanish*

Aquí, esto no lo pones
here this not CL.DO = put.2SG
‘Here, this, you’re not putting it.’

Other instances of cartographic-type structures will include cases of complementisers or subordination markers being followed by a topic or a *wh*-element, such as those in (26a), and, in Ibero-Romance, instances of quotative complementisers preceding *wh*-elements (26b). In Catalan, furthermore, interrogative complementisers can be *preceded* by topics (27):

(26) *Spanish*

- a. Me dijo **que el puzle** no lo quería comprar
 CL.IO = tell.PRFV.3SG that the puzzle not CL.DO = want.IMPV.3SG buy.INF
 ‘He/she told me that he/she didn’t want to buy the puzzle.’
- b. ¿**Que** **cuánto** te han costado estas bambas?
 that.QUOT how.much CL.IO = AUX.HAVE.3PL cost.PTCP these trainers
 ‘How much have you said these trainers have cost you!?’

(27) *Catalan*

- I **en Joan, que** no vindrà a sopar?
 and the John that.INT not come.FUT.3SG to dine.INF
 ‘Is John not coming to dinner?’

In Catalan and Spanish, Topic + *sí que* ‘yes that’ and *que sí que* ‘that yes that’ constructions, the former shared with Italian *sì che*, are also analysed (see (28)). The examples below are accompanied by the (adult-like) cartographic structures assumed in Villa-García & González Rodríguez (2020) (the label of XP is left open to debate, see p. 26). Although I remain agnostic as to exact implementation of their cartographic analysis, the key assumption that I adopt from this work and others below is that these constructions require more elaborate structure than a basic CP (in the CFC sense).

(28) *Spanish*

- a. **Pablo sí** que vino
 Paul yes that come.PRFV.3SG
 ‘Paul did come.’
- b. Creo que **sí** que ganaron la carrera
 think.1SG that yes that win.PRFV.3PL the race
 ‘I think that they did win the race’.

(29) $[_{\text{ForceP}} (\text{que}) [_{\text{TopicP}} [_{\text{XP}} \text{ sí } [_{\text{X}} \text{ que } [_{\text{TopicP}} [_{\text{FocusP}} [_{\Sigma\text{P}} [_{\text{TP}} \dots]]]]]]]]]]$

As far as Germanic languages are concerned, however, the V2-system makes a Split CP-domain substantially harder to detect and, indeed, not all approaches agree that the left periphery of Germanic V2-languages is “cartographic” in the traditional sense (cf. Giorgi & Pianesi 1997; Biberauer & Roberts 2015; Cormany 2015; Walkden 2017; Hsu 2017; 2021: for various views that assume a weakened left periphery in V2-languages, e.g., via conflation or bundled functional heads). However, limited instances of V3-orders are present in Germanic languages (these are more widespread in all colloquial and some vernacular varieties of German, although less so in Dutch; see Freywald et al. 2015, though cf. Meelen et al. 2020, on Dutch urban youth varieties).

(34) *Dutch*

Als het niet zo warm is, **dan** ga ik naar buiten
 when/if it not so hot be.3SG then go.1SG I to outside
 ‘When/if it isn’t so hot, then I’ll go out.’

The rationale for these Germanic diagnostics thus remains the same as for Romance: the structures involve two left-peripheral constituents and have been analysed as requiring an elaborate CP of some kind.

A final note regarding the status of V3-orders is in order. Given that V3-orders in Germanic, and Germanic V2 more generally, can be modelled in a non-cartographic manner (see Walkden 2017, and also references above), it is thus justified to wonder to what extent these V3-structures can properly be called “cartographic”. This raises issues about what these kinds of V3-orders are diagnosing in our context and its formal nature. As noted earlier, however, I simply take these Split CP-diagnostics (including V3-orders) to be diagnosing *further internal elaboration* within the CP-domain. The *exact richness* or *granularity* of this structural elaboration (e.g., if cartographic in the sense of Rizzi 1997; Cinque 1999; and subsequent work or “weaker”, as in Giorgi & Pianesi 1997; Cormany 2015; Walkden 2017, i.a.) is expected to be language-variant, and so secondary for our purposes so long as this granularity is greater than that of a CP with a single head. For this reason, I will continue to name these diagnostics “Split CP” or “cartographic-type” diagnostics, *not* “cartographic” diagnostics.

It is important to recognise that, firstly, the set of Split CP-structures introduced includes both structures that have been argued to be moved (e.g., wh-questions) vs. others that have received base-generation or adjunction analyses (e.g., Hanging Topics, V3 in Germanic; Grohmann 2000, den Dikken & Surányi 2017). Secondly, not all of them have been argued to support an articulated CP (e.g., Villa-García & Ott 2024). Other structures (e.g., recomplementation) have received both left-periphery-based and paratactic treatments (cf. Villa-García 2012a; Villa-García & Ott 2024). This is simply to acknowledge the breadth-first approach in my methodology: not all left fields with co-occurring constituents need imply an articulated CP under some analyses. Additionally, moved vs. base-generated left-peripheral constructions could follow different developmental pathways (Moscatti 2023). The lack of distinctions in this paper was due to a methodological consideration: maximising the data sample with two co-occurring CP-elements was prioritised here over differentiation across subtypes of Split CP-structures (e.g., by including only moved constituents in the latter case). In this paper, I explore the development of co-occurring left-peripheral elements in broad terms, and make the (preliminary) assumption that these may require articulated CPs (following, e.g., Haegeman & Greco 2016; Walkden 2017). By acknowledging their heterogeneous nature, I am therefore open to the possibility that these may indeed develop at different stages. As will be shown in section 3.4, however,

it is striking that all structures identified as Split-CP for the purposes of this study displayed the same developmental trends. Making subdivisions within these structures – and tracing their respective development – would nonetheless be a welcome future extension of this work, if sample sizes permit.

3.3 Summary of diagnostics, criteria for emergence and utterances excluded

Table 3 summarises all the left-peripheral structures searched for in every recording.

Functional domains	Structures analysed
CP-domain	• V-to-C movement (Germanic only)
	• Wh-questions
	• Yes/no questions (Germanic only)
	• Topicalisations and focalisations
	• Illocutionary complementisers (Romance only)
	• Finite embedding markers
Split CP	• Top > Wh (Romance only)
	• Top > Top/Foc (e.g., recursive topics; Romance only)
	• Comp > Wh/Top (Romance only)
	• QUOT <i>que</i> ‘that’ > Wh (Ibero-Romance only)
	• Top > INT <i>que</i> ‘that’ (Catalan only)
	• Topic + <i>sí que/sì che</i> ‘yes that’ and <i>que sí que</i> ‘that yes that’ structures (Romance and Ibero-Romance only, respectively)
	• Frame-setters, HTLD, CLD (Germanic only)
	• Conditional/temporal clauses with resumptive <i>dann/dan</i> ‘then’ (Germanic only)
	• Other possible combinations of the above

Table 3: Summary of structural diagnostics used.

For each diagnostic, I manually checked whether the structure appeared at least once in each file. A given functional domain (e.g., CP) is assumed to have emerged after the first attestation of a structure corresponding to that domain (e.g., CP or TP) and followed soon after (within ~2 files) by other uses of the same or other structures searched for in that domain. That is, isolated instances of a structure were not deemed sufficient evidence for emergence, and no conclusive statements were made in such cases – a “repeated uses” criterion for emergence. This criterion is necessarily somewhat arbitrary. There is no justified motivation for why ~2 files should be the relevant window of consideration. However, to avoid reaching theoretical conclusions on the

basis of isolated examples, it is methodologically important to adopt a conservative measure of emergence above (as done, for example, in Villa-García 2012b), *modulo* the fact that it is not fully satisfactory. In practice, however, “repeated uses” as per above and the more liberal metric of “first file of emergence” (Stromswold 1990) overlapped to a significant degree for this dataset, insofar as most first uses of a structure were almost always followed by further uses in subsequent files; this becomes clear in the tables in the Appendices, and is also discussed in passing at various points in section 3.4.

Additionally, depending on the recordings’ length for a given child, this criterion had to be weakened. This was only the case for the early-talker Spanish child Irene. Her recordings at the earliest stages are extremely brief, often including <50 words by the child, meaning they contain very few to no verbs and, therefore, often none of the structures analysed. It was also relaxed for the Split CP-diagnostics, which, by their very nature, are less frequent than the CP-structures out of which they are formed. This is crucial to ensure structures with rarer contexts of use are not underreported in the study. For a full breakdown of the counts obtained per file and structure, I refer the reader to Appendix A.

A quantitative analysis of all structures was also carried out, including an analysis of the variety in types for each structure (e.g., the types of wh-questions or wh-words used, verbs with which V2-order was found, among others). This was necessary to determine the productivity of the earliest CP-structures and to discard possible instances of rote-learned formulae. Full details can be found in Appendix B.

Identical repetitions of an adult utterance or the child’s previous utterance were excluded, as were lyrics and unintelligible utterances (either due to pronunciation or transcription).

3.4 Results

The analysis of the longitudinal data revealed a transparent order of emergence of the two sets of diagnostics. I begin by presenting an outline of the findings comparing the structures analysed as *groups* (CP vs. Split CP-structures). Then, I consider the relative order of emergence of the various CP-structures analysed. Overall, I introduce three theoretically consequential generalisations. These shed new light on the two theoretical topics introduced in sections 1 and 2 – directionality and granularity in development. Some theoretical commentary is offered, but the implications are discussed in section 4.

3.4.1 Order of emergence of structures and the role of MLUw

Based on the diagnostics used, the findings reflect two transparent stages, across the ten children. These are listed below, and exclude a period, in some children, where none of the structures are attested or, if they are, they represent isolated cases with unclear status.

- i. **Stage 1 – CP-structures only:** these typically included wh-questions, V2 (for Germanic), illocutionary complementisers (for Romance) and, more rarely, topicalisations/focalisations (typically MLUw $\approx$ 1.5). These structures become consolidated soon after their emergence.
- ii. **Stage 2 – Split CP-structures emerge:** Split CP-structures emerge well after the first CP-structures (in these children, generally at MLUw $\geq$ 2.5).

Stage 1 is generally long in duration (see **Table 12** later) and also includes the emergence of other structures, TP-elements (e.g., auxiliaries, set aside here; Appendix A and Bosch 2023a, for the full data). The emergence of Split CP-structures, however, comes systematically late, often *after* other complex syntactic configurations like embedding. To ease readability, I present the results of the corpus study via summarised tables, which indicate the point of emergence of these stages and the structures attested (per the criteria outlined earlier). **Table 4** summarises the data obtained for the Catalan, Italian and Spanish children and **Table 5** outlines the results for German and Dutch. CP-diagnostics are presented first, in blue, and then Split CP-diagnostics, in green. Cells which are not colour-shaded, but which feature a structure, indicate isolated, potentially unproductive utterances, per the emergence criterion above. Each table condenses the order of emergence of each of the CP-structures studied. The age/MLU ranges for each Stage are determined based on the first file in every child that satisfies the criteria for emergence for each domain, as defined above. In Germanic, where no wh-words are attested in a given period, but wh-less questions are present, this is noted as “wh-less”. The full tables are available in the Appendices.

Overall, MLUw proves to be a relatively stable metric, with the two stages emerging on average at the following MLUws. The average length (in months) of the first stage is also noted:

- **Stage 1:** MLUw 1.4 (range 1.09 – 1.96; average length of 10.5 months).
- **Stage 2:** MLUw 2.82 (range 2.32 – 3.57).

The analysis of the emergence of the various structures studied indicates that there exists a relationship between the MLUw of the child (understood as a metric of syntactic development) and the types of structures produced, with a clear relative order among those structures. The MLUw variances of all three stages also appear fairly homogenous (with the variances for Stages 1 and 2 being 0.0701 and 0.1755, respectively). Finally, a paired-samples t-test comparing the MLUw at the point of emergence of the two stages output statistically significant differences ($t(19) = 6.7188, p < .001$). This suggests Split CP-structures are produced at later stages, with significantly higher MLUw.

Language	Child	Age	MLUw	Wh-Q	Top/Foc	Illoc	Embed	Split CP	Stage
Catalan	Laura	1;07.20	1.03–1.15						
		1;10.22	1.15–1.3			✓			①
		2;02.13	1.3–1.88	✓		✓			
		2;08.30	1.88–2.42	✓	✓	✓			
		3;00.02	2.42–2.54	✓	✓	✓	✓		
		3;03.21	2.54 ¹⁰ –3.18	✓	✓	✓	✓	✓	②
	Gisela	1;07.14	1.04–1.12						
		1;08.24	1.13–1.58			✓			
		2;04.25	1.58–2.32			✓			①
		2;06.23	2.32–2.61	✓		✓			
		2;08.00	2.61–3.41	✓	✓	✓	✓	✓	②
Italian	Martina	1;07.18	1.26–1.57						
		1;08.02	1.57 ¹¹ –1.57	✓		✓			①
		1;08.17	1.57–1.99	✓	✓	✓			
		1;11.20	1.99–2.69	✓	✓	✓	✓		
		2;04.13	2.69–2.55	✓	✓	✓	✓	✓	②
	Rosa	1;07.13	1.27–1.75	✓					①
		2;04.23	1.75–2.6	✓	✓				
		2;06.29	2.6–2.87	✓	✓	✓			
		2;09.04	2.87–2.5 ¹²	✓	✓	✓	✓		
		2;10.14	2.5–3.24	✓	✓	✓	✓	✓	②
Spanish	Irene	0;11.01	1.42–1.32						
		1;04.16	1.32–1.88	✓					①
		1;08.09	1.88–2.95	✓	✓	✓			
		1;09.10	2.28–2.95	✓	✓	✓	✓		
		1;11.13	2.95–3.38	✓	✓	✓	✓	✓	②
	Koki	1;07.20	1.96–2.69		✓				①
		1;11.25	2.47–2.47	✓	✓		✓		
		2;02.27	2.47–2.69	✓	✓	✓	✓		
		2;04.18	2.69–3.38	✓	✓	✓	✓	✓	②

Table 4: Summary of production data by the Romance children.

¹⁰ The first Split CP-structure in Laura is attested in a file with MLUw 3.47. However, since the rest of the files in that period include substantially lower MLUs (around 2.5) and MLUw 3.0 is not reached until later files, I adopt the next file's MLU value as the value of emergence for Split CP-structures in Laura. These are the values cited in any future tables and summaries.

¹¹ The same approach as for Laura (see footnote 10) was adopted for the first CP-structure in Martina (attested at a file with MLUw 1.9, but surrounded by files at MLUw ~ 1.5).

¹² The first file of this period has a slightly higher MLUw than the final file, hence the apparently antichronological ordering. The same remark holds for other apparently antichronological timings.

Language	Child	Age	MLUw	V2	Wh-Q	Y/N-Q	Top/Foc	Embed	Split CP	Stage
German	Kerstin	1;03.22	1.09–1.36							
		1;05.24	1.36–1.28		✓					
		1;10.03	1.28–1.38		✓					①
		1;10.05	1.38–1.76	✓	✓					
		2;00.05	1.76–1.68	✓	✓		✓			
		2;00.10	1.68–2.13	✓	✓	✓	✓			
		2;07.23	2.13–2.32	✓	✓	✓	✓	✓		
		2;09.11	2.32–2.89	✓	✓	✓	✓	✓	✓	②
	Simone	1;09.11	1.54–1.62	✓						①
		1;10.20	1.62–2.31	✓	✓		✓			
		2;00.23	2.31–1.96	✓	✓	✓	✓			
		2;04.20	1.96–2.78	✓	✓	✓	✓	✓		
		2;06.23	2.78–3.46	✓	✓	✓	✓	✓	✓	②
Dutch	Josse	2;00.07	1.2–1.94	✓	✓					①
		2;03.28	1.94–2.42	✓	✓	✓	✓			
		2;09.02	2.42–3.57	✓	✓	✓	✓	✓		
		2;11.09	3.57–3.2	✓	✓	✓	✓	✓	✓	②
	Sarah	1;06.16	1.12–1.17							
		1;07.21	1.17–1.09		Wh-less					
		1;10.05	1.09–1.68	✓	Wh-less					①
		2;00.17	1.68–2.11	✓	Wh-less	✓	✓			
		2;02.18	2.11–3.52	✓	✓	✓	✓			
		3;00.19	3.52–4.92	✓	✓	✓	✓	✓	✓	②

Table 5: Summary of production data by the Germanic children.

Crucially, age is not a reliable predictive factor. **Table 6** displays the age of emergence of the three sets of structures analysed. Age of emergence presents high variance. The variances for each variable (Stage 1 and 2) when measured in terms of age (in months) vary substantially (being 10.4556 and 17.25, respectively).

The age at which children produce specific structures can thus vary significantly from child to child. There is vast variation: at 2;00, Josse is still at Stage 1, while Irene reaches Stage 3 at 1;11. Similarly, Rosa and Laura reach Stage 2 by the same age (2;04), but Laura reaches Stage 3 around half a year later (2;10 vs. 3;03). The most obvious outlier is Irene, who reaches Stage 3 by 1;11, well before several children reach Stage 1. This echoes previous literature showing high variability in age of acquisition (Paradis & Genesee 1997; Caprin & Guasti 2009; Friedmann et al. 2021). These results further endorse the need to make crosslinguistic comparisons with another

	Stage 1	Stage 2
Laura	1;10.22	3;03.21
Gisela	2;04.25	2;08.00
Martina	1;08.02	2;04.13
Rosa	1;07.13	2;10.14
Irene	1;04.16	1;11.13
Koki	1;07.20	2;04.18
Kerstin	1;10.03	2;09.11
Simone	1;09.11	2;06.23
Josse	2;00.07	2;11.09
Sarah	1;10.05	3;00.19

Table 6: Age of emergence across the two stages.

less variable metric, such as MLUw, or centre the attention on stages observed, rather than ages (Friedmann & Reznick 2021: 1). Guttman Scales have proven a useful heuristic tool in this latter respect; I incorporate them in the following section.

3.4.2 The development of CP-structures (Stage 1): an early phenomenon

I now go on to unpack the broad observations above, outlining first the specific patterns of acquisition observed at each Stage and, then, introducing three theoretically consequential generalisations.

First, Stage 1 suggests that CP might be acquired early. I provide evidence from various diagnostics that shows this conclusion is likely corroborated.

As for Romance, the earliest recordings contain *wh*-questions, illocutionary complementisers and some topics/foci. Some examples of these early CP-structures are given below:

(35) *Catalan*

- a. **Què** és? (Laura, MLUw 1.3)
what be.3SG
‘What is it?’
- b. Ai, **que** crema! (Laura, MLUw 1.35)
ouch that.EXCL burn.3SG
‘Ouch, it’s burning!’
- c. **Que** cau! (Laura, MLUw 1.3)
that.EXCL fall.3SG
‘It’s falling!’

(36) *Italian*

- a. **Dov'**è? (Rosa, MLUw 1.39)
 where-be.3SG
 'Where is it?'
- b. **Pello** oio io [: quello voglio io] (Rosa, MLUw 1.75)
 that want.1SG I
 'That, I want.'
- c. **Ove** va chetto? (Martina, MLUw 1.57)
 where go.3SG this
 'Where does this one go?'
- d. **Che** gira (Martina, MLUw 1.9)
 that.EXCL stir.3SG
 'He/she stirs (it).'

(37) *Spanish*

- a. ¿**Quién** es? (Irene, MLUw 1.32)
 who be.3SG
 'Who is it?'
- b. ¿**Qué** hay ahí mamá? (Irene, MLUw 1.69)
 what have.3SG there mum
 'What is there mum?'

German and Dutch children likewise present evidence of CP-structures at this stage, and to a greater extent than Romance children. Firstly, there is evidence of early wh-questions and some topicalisations (of both adverbs and objects), as shown in (39).

(38) *Dutch*

- a. **Wo** 's den Nina? (Kerstin, MLUw 1.28)
 where be.3SG the.ACC Nina
 'Where is Nina?'
- b. **Wat** doet ie nou? (Josse, MLUw 1.59)
 what do.3SG he now
 'What is he doing now?'

(39) *German*

- a. **Jetzt** kommt ein Affe (Kerstin, MLUw 1.76)
 now come.3SG a monkey
 'Now a monkey comes.'

- b. **Blume** eßt nicht (Simone, MLUw 1.69)
 flower eat.3SG not
 ‘Flower, he/she doesn’t eat.’
- c. **Bau(e)n** will ich (Simone, MLUw 1.71)
 build.INF want.1SG I
 ‘To build (something), I want.’

There are, also, noticeable signs of a distinction between non-finite verbs (which remain in-situ and yield OV-order where an object is present) and finite verbs (which can be shown to have raised outside the *vP*).¹³

(40) *German*

- a. Ich will de(r) Ball (Kerstin, MLUw 1.58)
 I want the ball
 ‘I want the ball.’
- b. Wieder aufräumen (Kerstin, MLUw 1.34)
 again up.tidy.INF
 ‘To tidy up again.’
- c. Ma(ch) ma(l) Schuh ab (Kerstin, MLUw 1.53)
 make.IMP.2SG once shoe off
 ‘Take off (the) shoe.’
- d. Mag nicht Küche(n) backe(n) (Simone, MLUw 1.62)
 like.1SG not cakes bake.INF
 ‘I don’t like to bake cakes.’
- e. Bluse auszieh(e)n (Simone, MLUw 1.62)
 blouse off.take.INF
 ‘To take off (the) blouse.’

(41) *Dutch*

- a. Komt ie (Josse, MLUw 1.2)
 come.3SG he
 ‘He comes.’
- b. Koffie drinken (Josse, MLUw 1.44)
 coffee drink.INF
 ‘To drink coffee.’

¹³ Note that several examples feature Root Infinitives (RIs). It is beyond the scope of this work to provide an analysis of RIs crosslinguistically. Whenever finite verbs are used, though, they are almost without exception located in V2 position (Appendix B).

- c. Doet niet meer (Josse, MLUw 1.55)
do.3SG not more
'He/she doesn't do (it) anymore.'
- d. I(k)moe(t) ete(n) (Sarah, MLUw 1.16)
I-must.1SG eat.INF
'I have to eat.'
- e. I(k)wi(l) farkj [: varkentje] (Sarah, MLUw 1.25)
I-want.1SG pig.DIM
'I want piglet.'
- f. Boekje lezen (Sarah, MLUw 1.68)
book.DIM read.INF
'To read book/booklet.'

Finally, at the end of this stage finite embedding markers also emerge for all children. Some instances are given in (42)–(46):

- (42) *Catalan*
Saps **que** no vindrà ningú? (Gisela, MLUw 2.61)
know.2SG that not come.FUT.3SG no.one
'Do you know no one is going to come?'
- (43) *Italian*
È l'omo **che** ccrive (Martina, MLUw 2.55)
be.3SG the-man that write.3SG
'It/he is the man that writes.'
- (44) *Spanish*
Deja esto ahí **para que** no se rompe (Koki, MLUw 3.08)
leave.IMP.2SG this there so that not CL.REFL = break.3SG
'Leave this there so that it doesn't break.'
- (45) *German*
Weil der Tommy gebißen hat (Simone, MLUw 2.52)
because the Tommy bite.PTCP AUX.HAVE.3SG
'Because Tommy bit (something/someone).'
- (46) *Dutch*
Ik krijg deze **a(l)s** ik uit de crèche ben (Sarah, MLUw 3.52)
I get.1SG this when I out the nursery be.1SG
'I get this when I am out of the nursery.'

Age	MLU	Wh-Q	Top/Foc	Illoc	Embed	Split CP
1;07.20	1.03					
1;09.07	1.09					
1;10.22	1.15			✓		
1;11.12	1.15			✓		
2;02.05	1.35	✓				
2;02.13	1.3	✓				
2;04.11	1.44	✓				
2;05.08	1.64					
2;06.25	1.76	✓				
2;07.20	1.78	✓		✓		
2;08.30	1.88	✓	✓	✓		
2;11.17	1.98	✓	✓	✓		
3;00.02	2.42	✓	✓	✓	✓	
3;03.21	3.47	✓	✓	✓	✓	✓
3;05.13	2.54	✓	✓	✓	✓	✓
3;10.00	2.97	✓	✓	✓	✓	✓
3;10.01	2.91	✓	✓	✓	✓	✓
3;11.12	3.0	✓	✓	✓	✓	✓
4;00.10	3.18	✓	✓	✓	✓	✓

Table 7: Production of structures by Laura.

I substantiate this with **Tables 7** and **8**. Due to space considerations, only two children are given, but the patterns recur across the ten children (Appendix A).

A summary of the types and tokens of all CP-structures attested throughout Stage 1 is given in **Table 9**. The first value in parentheses next to the number of wh-questions indicates the proportion of copular wh-questions and the second one in German and Dutch signals the proportion of wh-less questions. This illustrates the variety of predicates observed with wh-questions (the variety in wh-words is available in Appendix B). For simplicity, only attestation of V2 is indicated below (with a check mark); for a full quantitative break-down of V2 and V-final orders, see Appendix B.

Given this data, it is reasonable to suggest that productive knowledge of structures like wh-questions, V2, topics and illocutionary complementisers is in place very early on. CP-structures are abundant from the start of Stage 1, with multiple structures instantiated across recordings, as shown by **Table 9**. Importantly, the knowledge of CP and V2 systems established at Stage 1 involves at most a basic CP. There is virtually no production data supporting the availability

Age	MLU	V2	Wh-Q	Y/N-Q	Top/Foc	Embed	Split CP
1;06.16	1.12						
1;07.21	1.17		Wh-less				
1;08.28	1.07						
1;09.10	1.17		Wh-less				
1;10.05	1.09	✓					
1;10.13	1.17	✓					
1;11.01	1.25	✓					
1;11.15	1.37	✓	Wh-less				
2;00.17	1.68	✓		✓	✓		
2;01.10	1.88	✓		✓	✓		
2;02.18	2.11	✓	✓	✓			
2;03.16	2.05	✓	✓	✓			
2;04.02	2.53	✓	✓	✓	✓		
2;04.09	2.34	✓	✓	✓			
2;04.27	2.46	✓	✓	✓	✓		
2;05.09	2.47	✓	✓	✓	✓		
2;05.22	2.59	✓	✓	✓	✓		
2;06.04	2.74	✓	✓	✓	✓		
2;06.11	2.45	✓		✓	✓		
2;06.18	2.8	✓	Wh-less	✓	✓		
2;07.16	2.51	✓	✓	✓	✓		
2;08.06	2.66	✓	✓	✓	✓		
2;08.19	2.97	✓	✓	✓	✓		
2;09.02	2.59	✓	✓	✓	✓		
2;09.07	3.15	✓	✓	✓	✓		
2;10.18	2.88	✓	✓	✓	✓		
2;11.03	2.87	✓	✓	✓	✓		
2;11.27	3.64	✓	✓	✓	✓		
3;00.19	3.52	✓	✓	✓	✓	✓	✓
3;01.17	3.06	✓	✓	✓	✓	✓	✓
3;02.13	3.82		✓	✓	✓	✓	✓
3;03.21	3.05	✓	✓	✓	✓	✓	✓
3;04.13	3.15	✓	✓	✓	✓	✓	✓
3;05.30	2.89	✓	✓	✓	✓	✓	✓
3;05.30	2.89	✓	✓	✓	✓	✓	✓
3;07.25	3.24	✓	✓	✓	✓	✓	
3;10.07	3.71	✓	✓	✓	✓	✓	✓
3;11.04	4.07	✓	✓	✓	✓	✓	✓

(Contd.)

Age	MLU	V2	Wh-Q	Y/N-Q	Top/Foc	Embed	Split CP
4;00.11	3.81	✓	✓	✓	✓	✓	✓
4;00.30	4.08	✓	✓	✓	✓	✓	
4;01.11	4.66	✓	✓	✓	✓	✓	✓
4;03.04	5.37	✓	✓	✓	✓	✓	✓
4;04.28	4.28	✓	✓	✓	✓	✓	
4;05.29	4.7	✓	✓	✓	✓	✓	✓
4;06.12	5.06	✓	✓	✓	✓	✓	✓
4;07.25	4.62	✓	✓	✓	✓	✓	✓
4;08.03	5.03	✓	✓	✓	✓	✓	✓
4;09.13	6.07	✓	✓	✓	✓	✓	✓
4;09.29	5.2	✓	✓	✓	✓	✓	✓
4;11.15	4.01	✓	✓	✓	✓	✓	✓
5;02.13	4.92	✓	✓	✓	✓	✓	✓

Table 8: Production of structures by Sarah.

	V2	Wh-Q	Y/N-Q	Top/Foc	Illoc	Embed	Length
Laura		15 (9)		4	42	4	1;10.22–3;03.21 (MLUw 1.15–2.54)
Gisela		1 (0)		0	6	0	2;04.25–2;08.00 (MLUw 1.58–2.61)
Martina		21 (16) ¹⁴		3	7	8	1;08.02–2;04.13 (MLUw 1.57–2.69)
Rosa		133 (131)		12	3	8	1;07.13–2;10.14 (MLUw 1.27–2.5)
Irene		18 (6)		3	10	4	1;04.16–1;11.13 (MLUw 1.32–2.95)
Koki		32 (27)		7	2	4	1;07.20–2;04.18 (MLUw 1.96–2.69)
Kerstin	✓	16 (14, 2) ¹⁵	21	27		1	1;10.03–2;09.11 (MLUw 1.28–2.32)
Simone	✓	166 (154, 7)	3	105		24	1;10.03–2;06.23 (MLUw 1.54–2.78)
Josse	✓	62 (43, 25) ¹⁶	37	68		1	2;00.07–2;11.09 (MLUw 1.2–3.57)
Sarah	✓	124 (100, 46)	104	116		0	1;10.05–3;00.19 (MLUw 1.09–3.52)

Table 9: CP-structures produced at Stage 1 and its length.

¹⁴ 2 wh-questions are non-finite.

¹⁵ 1 wh-question is non-finite.

¹⁶ As for footnote 15.

of an internally elaborate CP. The specific articulation of CP, including the specific V2 system (e.g., ‘Force-V2’ or ‘Fin-V2’ in Poletto’s 2002, and Wolfe’s 2015 typology; see also Biberauer & Roberts 2015) is expected to be consolidated at Stage 2, when cartographic-type structures appear. It follows, too, that V3-configurations requiring finer-grained cartographic-type structures (e.g., those introduced in section 3.2.2) are expected not to emerge until Stage 3 – an expectation which is borne out and will be discussed in section 3.4.3.

This is, therefore, Generalisation 1: CP is an early phenomenon in this data.

Generalisation 1 (Early Acquisition of CP). CP-structures emerge early on in the developmental data.

This generalisation poses a challenge to bottom-up maturation,¹⁷ but is coherent with all inward-growing, continuity and neo-emergentist approaches (see section 4). This first result, then, aligns with the fundamental shared assumption in these frameworks – (some) CP knowledge develops early.

This conclusion is further underscored once the development of several left peripheral structures is examined more carefully. We noted above that embedding markers emerge late, across all children. This stands in contrast with the rest of the CP-diagnostics studied. Implicit in the above is, thus, another important result, echoed by at least two separate empirical points. Structurally very high elements emerge early on in several children; this is the case of topics and illocutionary complementisers.

First, the (near-)simultaneous emergence of embedding markers and topicalisation in Friedmann et al. (2021) (their Stage 3) is, for many children, not replicated. At most, it only holds for some of the Romance children. Topicalisation emerges considerably before finite embedding in several of the children studied (**Table 10**). The difference between topics and embedding is particularly stark in Germanic children, who use topics from the earliest files and MLUw ranges. The MLUw difference between topics and embedding is highly statistically significant, according to a paired-samples t-test ($t(9) = 3.3386, p = .009$). Note that, in order to facilitate a comparison with Friedmann et al.’s (2021) conjecture (which bears on topicalisation and embedding, *not* the left-peripherally lower focalisation), this analysis excluded fronted objects in Romance that are possible instances of foci or that are ambiguous between focalisation/topicalisation.¹⁸

¹⁷ Although one could technically maintain that Generalisation 1 is only problematic for bottom-up approaches if CP emerges *before* TP, as an anonymous reviewer notes. This relative interpretation is possible (and is more explicitly endorsed in Friedmann et al. 2021). However, Generalisation 1 emphasises that the data would still force a departure from the original spirit of bottom-up approaches (e.g., Radford 1990; Rizzi 1993) where CP was explicitly predicted to emerge “*late*” (a term loosely defined, but often correlated with age 2–2.5 in Radford 1990). See also Bosch (2023a) and Heim & Wiltschko (2025), who provide some evidence that CP-structures are as early as, if not earlier, than TP-structures.

¹⁸ However, this could be an underestimate of the occurrence of topics and does not control for a scenario where topicalisations superficially emerge later than focalisations simply because they omit clitics in CLLD (see Guasti 1993;

	Topicalisation	Embedding
Laura	2;11.17	3;00.02
	1.98 MLUw	2.42 MLUw
Gisela	2;08.00	2;08.00 (same file)
	2.61 MLUw	2.61 MLUw
Martina	1;08.17	1;11.20
	1.56 MLUw	1.99 MLUw
Rosa	2;04.29	2;06.29
	1.77 MLUw	2.6 MLUw
Irene	1;08.09b	1;09.10
	2.24 MLUw	3.28 MLUw
Koki	1;11.25	1;11.25 (same file)
	2.47 MLUw	2.47 MLUw
Kerstin	2;00.05	2;07.23
	1.76 MLUw	2.13 MLUw
Simone	1;10.20	2;04.20
	1.62 MLUw	1.96 MLUw
Josse	2;03.28	2;09.02
	1.94 MLUw	2.42 MLUw
Sarah	2;00.17	3;00.19
	1.68 MLUw	3.52 MLUw
Average	1.96 MLUw	2.54 MLUw

Table 10: Emergence of topicalisation vs. embedding markers.

Examples (47)–(49) illustrate some of the early utterances containing topics, in Romance and German, some repeated from (39).

(47) *Italian*

- a. **Ove** va chetto? (Martina, MLUw 1.57)
 where go.3SG this
 ‘Where does this one go?’
- b. **Bere** voglio (Martina, MLUw 1.66)
 drink.INF want.1SG
 ‘To drink, I want.’
- c. **Que’** lo l’apri? (Rosa, MLUw 1.78)
 that CL.DO = CL.DO = open.2SG

(48) *German*

- a. **Jetzt** kommt ein Affe (Kerstin, MLUw 1.76)
 now come.3SG a monkey
 ‘Now a monkey comes.’
- b. **Bau(e)n** will ich (Simone, MLUw 1.71)
 build.INF want.1SG I
 ‘To build (something), I want.’

(49) *Dutch*

- Daar** gaat ie (Josse, MLUw 2.14)
 there go.3SG he
 ‘There he goes.’

Second, illocutionary complementisers also consistently emerge very early: Laura, Gisela, Irene and Koki display illocutionary complementisers at MLUw 1.15, 1.58, 1.88 and 1.96, respectively, and from the earliest files. Some examples of early illocutionary complementisers are given in (50)–(51). Importantly, this pattern is not exclusive to these children or languages, but was already supported with bigger Catalan/Spanish samples in Bosch (2023b) and for Romanian in Stoicescu (2025).

(50) *Catalan*

- a. Ai, **que** crema! (Laura, MLUw 1.35)
 ouch that.EXCL burn.3SG
 ‘Ouch, it’s burning!’
- b. **Que** cau! (Laura, MLUw 1.3)
 that.EXCL fall.3SG
 ‘It’s falling!’
- c. **Que** no la puc treure (Gisela, MLUw 2.32)
 that.QUOT not CL.DO = can.1SG take.out.INF
 ‘(I’ve said) I can’t take it out.’

(51) *Spanish*

- Que** rompió (Irene, MLUw 1.88)
 that.QUOT break.3SG.PST
 ‘(What happened is that) it broke / he broke it.’

Recall that these main-clause complementisers are typically assumed to be located high in the CP-domain (often in Rizzi’s Force head) or often in even higher Speech Act heads (section 3.2.1). Illocutionary complementisers, then, bring a new, understudied piece of data suggesting structural height does not always correlate with late acquisition.

In sum, this brings us to the second generalisation we will consider. The long-held assumption that acquisition “recapitulates” a UG-given spine *cannot* be maintained, at least in its stronger, cartographic form. Acquisition and structural height can, and in fact *do*, mismatch in the present data (Generalisation 2) (from Bosch & Biberauer 2024: 109):

Generalisation 2 (Structural Height and Acquisition Mismatch). There is a dissociation between structural height and order of emergence. Acquisition does not proceed successively upwards; some syntactically very high elements emerge early.

This mismatch creates a *Directionality Problem*: if a cartographic-type spine is assumed, we cannot account for acquisition patterns with *directionality* of maturation as our only developmental driving factor. Regardless of the directionality assumed, we predict some structures to be late-acquired, only for them to emerge very early (we take this up again in section 4). Concretely, Generalisation 2 undermines in particular cartographic bottom-up approaches such as the Growing Trees Hypothesis (Friedmann et al. 2021) as *universal* pathways: insofar as both topics and illocutionary complementisers are hosted at the top of a cartographic left periphery (e.g., TopicP in the former case, and ForceP or higher Speech-Act heads in the latter), both elements should emerge at the end of the learning path (Friedmann et al.’s 2021 Stage 3). The opposite pattern is observed.

Now, the above does not necessarily imply that these developmental patterns will always replicate themselves in other languages or children (a separate empirical question). It remains possible that some structurally high elements, like topics, emerge late with embedding in other languages, as Friedmann et al. (2021) show for Hebrew. What it *does* imply, however, is that topics and illocutionary complementisers do not *universally* emerge alongside other (late-acquired) structurally high elements across all languages and children. Additionally, it is worth noting that the timing discrepancies observed across the two language families in **Table 10** are also telling. The timing of topics in Romance is fairly variable across children, but is systematically early in Germanic. This is suggestive in light of literature pointing to Germanic topicalisation being formally much simpler than the options available in (some) Romance varieties (see Biberauer & Roberts 2015: 15–18, for an explicit comment), aligning with the earlier emergence of topics in the former.¹⁹ This tentative correlation between formal complexity of topicalisation and its age of acquisition was actually corroborated by Bosch & Biberauer (2025), who show for a sample of 10 languages that early topic acquisition systematically correlates with languages whose

¹⁹ At this point, one might raise an alternative interpretation of the patterns in Generalisation 2; namely, that these “structurally high” elements do not involve structurally high heads *at this particular stage*, but instead recruit other lower heads like Fin until those higher heads mature. I discard this option on parsimony grounds. There is no clear motivation at present (except theory-internal reasons) for rethinking these structures as reflecting the development of left-peripherally low heads (e.g., Fin). Additionally, introducing “misanalysis” in the account would significantly hamper the falsifiability of existing cartographic proposals.

topicalisation displays either adjunction/base-generation (e.g., French) or ‘pure’ A’-movement (e.g., European Portuguese; following van Urk 2015; Cinque 1990, i.a.). Languages with topics displaying a *mixed* set of A/A’-properties (e.g., CLLD languages, but also clitic-less topicalisation in Brazilian Portuguese and Hebrew) are late acquired, without exception. Their results again underscore the L1-specificity of the acquisition of topicalisation.

Thus far we have established that there is variation in the emergence of structures located at the top of the left periphery; some emerge early (e.g., illocutionary complementisers, topics), others do not (embedding), with some crosslinguistic differences. I have argued that this variation is consequential, questioning an absolute ban on early production of structurally high elements. Setting variation aside, an additional question that emerges from the previous two sections, however, is if there are any shared pathways of acquisition of the various CP-structures observed. That is, even after establishing problems with directionality-only approaches, what universality is left in learning paths? I now incorporate Guttman Scales to address this point.

Guttman Scales have often been harnessed to reveal shared temporal orders in language acquisition (Barton 1976; Levelt et al. 1999/2000; Fikkert & Levelt 2008; Friedmann & Reznick 2021; Friedmann et al. 2021; van ’t Veer 2023). These scales are hypotheses about the structure of a dataset with respect to an attribute. This type of cumulative scaling can help establish an ordinal scale with an arrangement such that, if a structure X has emerged (e.g., by a given time), any other structures that fall below it in the scale will have also been acquired by that point. **Table 11** summarises the relative order of emergence of each of the CP-structures analysed (organised from earliest to latest, left to right).

Child	Order of emergence
Laura	Illoc > Wh > Topic > Embed > Split CP
Gisela	Illoc > Wh > Topic/Embed/Split CP
Martina	Illoc/Wh > Topic > Embed > Split CP
Rosa	Wh > Topic > Embed > Illoc > Split CP
Irene	Wh > Illoc/Topic > Embed > Split CP
Koki	Topic > Wh/Embed > Illoc > Split CP
Kerstin	Wh > V2 > Topic/YN > Embed > Split CP
Simone	V2 > Wh/Topic > YN > Embed > Split CP
Josse	Wh/V2 > Topic/YN > Embed > Split CP
Sarah	V2 > Y/N/Topic > Wh > Embed/Split CP

Table 11: Relative order of emergence of diagnostics studied.

Crucially, the orderings in **Table 11** suggest that extremely few individual CP-structures can be ordered into a universal scale. The only, very broad, emergence scales that emerge for the two language families are the ones below. Note that illocutionary complementisers are the only structures that cannot be ordered relative to other structures; they quite often emerge from the start (Laura, Gisela, Martina), but also a bit later (Koki, Rosa).

- **Romance:** Wh/Topic > Embed > Split CP
- **Germanic:** Wh/V2/Topic/YN > Embed > Split CP

Here, only these much broader scales can be obtained from this data, which at most encode the observation that CP-structures emerge before embedding, which in turn emerges before Split CP-structures. No matter which order we try to establish for other structures (e.g., Wh > Illoc, or Wh > Topic), then, we almost always cannot get a universal one, some child/language will show a different order of emergence. This contrasts with the results observed in Friedmann et al. (2021) for Hebrew children and Meira & Grolla (2022) for Brazilian Portuguese, who argue that structurally higher elements (e.g., topics, embedding) generally emerge significantly later than lower ones (e.g., wh-questions).

This outcome is suggestive, and endorses Generalisation 2 further: if acquisition and structural height often mismatch, then we might expect shared Guttman Scales to be hard to establish. In section 4, I propose to reconceptualise this aspect of development and argue that *granularity* appears to bear a potentially more consequential role.²⁰

Thus, Generalisations 1 and 2 reinforce the assumption that CP is acquired early (section 2), but provide novel support against strict directionality as recently conceived of in cartographic approaches. This confers a clear advantage to continuity, inward-growing and neo-emergentist approaches alike. Having addressed *directionality* in development so far, I now introduce the final set of results (Stage 2), which instead concerns *granularity*. I argue in section 4 that this stage singles out neo-emergentist approaches as better suited for this data.

²⁰ An anonymous reviewer points out that Generalisation 2 is only incompatible with Growing Trees regarding the timing of illocutionary complementisers; wh-questions generally (though not always) emerge before topics in **Table 11**, in line with Growing Trees' Stage 2 and 3. Adopting this reasoning, however, risks being stipulative: illocutionary complementisers are generally treated as high CP-elements in cartographic approaches (see section 3.2), and have been reported to be early acquired for all of Catalan, Spanish, some Italian children (this paper; Bosch 2023b) and Romanian (Stoicescu 2025). All other things being equal, this pattern alone should be taken to contradict a Growing Trees approach. Further, the extent to which the acquisition of topicalisation follows a Growing Trees pathway crosslinguistically is far from straightforward: I refer the reader to Bosch & Biberauer (2025), who show that early topics are observed in a significant range of languages, correlating with featural complexity; to de Cat (2007; 2009), who reports early topics for French, even *before* wh-questions are acquired; and to Villa-García & Snyder (2009), who compare the emergence of lexical (topical) subjects with wh-questions in Spanish.

3.4.3 Split CP-structures (Stage 2): refining the system

Generalisations 1–2 provide new data and theoretical synthesis corroborating already-existing results; notably, previous work supporting early acquisition of CP (section 2). However, it is at Stage 2 that a novel empirical generalisation emerges. At this stage, evidence for co-occurrence of multiple CP-elements emerges robustly, something that was virtually absent up until this point.

I will argue that this generalisation provides fresh insights into the development of the left periphery and functional categories more broadly – by emphasising an underexplored aspect of child grammatical knowledge, its syntactic *granularity*. This is discussed in section 4; I focus first on its empirical manifestation. Fundamentally, Generalisation 3 uncovers an apparent discrepancy in the quality and quantity of CP-related productions in the child data at various stages. **Table 12** shows the clear “late” nature of Split CP-structures vis-à-vis CP-structures.

	CP-structures	Split CP-structures
Laura	1;10.22	3;03.21
	1.15 MLUw	2.54 MLUw
Gisela	2;04.25	2;08.00
	1.58 MLUw	2.61 MLUw
Martina	1;08.02	2;04.13
	1.57 MLUw	2.69 MLUw
Rosa	1;07.13	2;10.14
	1.27 MLUw	2.5 MLUw
Irene	1;04.16	1;11.13
	1.32 MLUw	2.95 MLUw
Koki	1;07.20	2;04.18
	1.96 MLUw	2.69 MLUw
Kerstin	1;10.03	2;09.11
	1.28 MLUw	2.32 MLUw
Simone	1;09.11	2;06.23
	1.54 MLUw	2.78 MLUw
Josse	2;00.07	2;11.09
	1.2 MLUw	3.57 MLUw
Sarah	1;10.05	3;00.19
	1.09 MLUw	3.52 MLUw

Table 12: Emergence of CP vs. Split CP-structures.

Some structures with multiple elements in the left periphery attested at this stage in Catalan, Italian and Spanish are shown in (52)–(54). A very common kind of structure displaying evidence for a Split CP involves topics preceding *wh*-questions (52a), (53a), (53b), (54b). Example (52b) includes recursive topics. (54a) presents a quotative complementiser preceding a *wh*-word.

(52) *Catalan*

- a. I **el ioc’ioc**, **què** fa? (Laura, MLUw 2.91)
 and the ioc-ioc what do.3SG
 ‘And the ioc-ioc (onomatopoeia), what does it do?’
- b. **Ara**, **aquest**, no puc (Gisela, MLUw 2.71)
 now this no can.1SG
 ‘Now, this one, I can’t.’

(53) *Italian*

- a. **Quetto**, **cosa** fa? (Martina, MLUw 2.37)
 this what do.3SG
 ‘This, what does it do?’
- b. **Questo**, **dove** si mette? (Rosa, MLUw 2.5)
 this where CL.REFL = put.3SG
 ‘This, where do you put it?’

(54) *Spanish*

- a. ¿**Que** **qué** nos falta? (Irene, MLUw 3.23)
 that.QUOT what CL.IO = lack.3SG
 ‘(I’ve asked) What are we missing?’
- b. ¿**Esas bonitas cortinas**, **dónde** las compraron? (Koki, MLUw 2.71)
 those beautiful curtains where CL.DO = buy.PRFX.3PL
 ‘Those beautiful curtains, where did they buy them?’

At Stage 3, at least three separate slots are also detectable in the production data. Example (55a) combines what appears to be an adverbial topic *ahir*, followed by a second topic/focus, alongside another topic (the CLLD topic *a jo*). Similarly, example (55b) displays a *que sí que* structure (see section 3.2.2) and (56a)–(56b) instantiate two topics (*ahora* and *ésto/todos los muchachos*) preceding a *wh*-element ((a) *dónde*). By MLUw ≥ 2.5 I conclude, then, that all children can harness two, and, based on some examples, seemingly also three, separate projections within CP.

(55) *Catalan*

- a. **Ahir, tu, a jo,** me vas dir que me (Laura, MLUw 2.91)
 yesterday you to me CL.IO = AUX.GO.2SG tell that CL.IO =
 l'explicaries avui
 CL.DO = explain.COND.2SG today
 'Yesterday, YOU told ME that you would explain it to me today.'
- b. **Que sí que** vol (Gisela, MLUw 2.61)
 that.QUOT yes that.QUOT want.3SG
 '(I've said) He/she does want to.'

(56) *Spanish*

- a. ¿Y **ahora, ésto, dónde** se mete? (Irene, MLUw 4.01)
 and now this where CL.IMPRS = put.3SG
 'Now, this, where does it go?'
- b. ¿**Ahora, todos los muchachos, a dónde** se van (Koki, MLUw 3.38)
 now all the boys to where CL.REFL = go.3PL
 a ir?
 to go.INF
 'Now, all the boys, where are they going to go?'

In Germanic languages, Stage 3 involves the emergence of the restricted kinds of V3-orders discussed in section 3.2.2. Frame-setters yielding V3-orders are illustrated in (57a)–(58c). (57a) and (58a) contain instances of Hanging Topics, while (57b), (57c), (58b), (58c) present cases of Contrastive Left-Dislocation.

(57) *German*

- a. Und **dann das** brauch(e) ich (Kerstin, MLUw 2.68)
 and then that need.1SG I
 'And then I need that.'
- b. **Den Teddy,** ich halt **den** so (Simone, MLUw 2.78)
 the.ACC Teddy I hold.1SG PRON.ACC so
 'The Teddy, I hold him like this.'
- c. **Die Leute, die** ham [: haben] des (Simone, MLUw 4.89)
 the people they have.3PL that
 puttemacht [: kaputtgemacht]
 break.make.PTCP
 'The people, THEY have broken that.'

(58) *Dutch*

- a. En **toen daar** was Hanneke weggegaan (Josse, MLUw 3.91)
 and then there AUX.BE.PST.3SG Hanneke away.go.PTCP
 ‘And then, there Hanneke had left.’
- b. **De staart, die** had je goed gevonden (Sarah, MLUw 3.15)
 the tail that AUX.HAVE.PST.2SG you good find.PTCP
 ‘The tail, that you found good.’
- c. **Andere boekje, die** lees ik nooit (Sarah, MLUw 2.89)
 other book.DIM that read.1SG I never
 ‘The other book/booklet, I never read (it).’

Finally, cases of conditional/temporal clauses with adverbial resumption are exemplified below. These are attested primarily in the Dutch data and are abundant in both Sarah and Josse after MLUw 2.5 (36 and 12 instances are recorded, respectively). 1 clear-cut example is found in the German data (in Simone).

(59) *Dutch*, V3-order in conditional/temporal clauses with resumptive *dann/dan*

- a. **Als** ik naar de grote school gaat, **dan** komt (Josse, MLUw 3.78)
 when/if I to the big school go.3SG then come.3SG
 sneeuw
 snow
 ‘When I go to the big school, then snow comes.’
- b. **Als** je drie bent, **dan** mag je niet van de (Sarah, MLUw 3.05)
 when/if you three be.2SG then may.2SG you not from the
 politie op school
 police in school
 ‘When/if you’re three, the police don’t let you into school.’

(60) *German*

- Wenn** du keine Suppe aufißt, **dann** hau ich (Simone, MLUw 4.89)
 when/if you no soup up.eat.3SG then smack I
 den dir
 PRON.ACC you.DAT
 ‘If you don’t eat up the soup, I’ll smack you.’

The data moreover indicates not just a delay, but *also* abruptness and “explosiveness” in the way these structures emerge. For the vast majority of children no Split CP-structures or extremely few are detectable before MLUw ~2.5; they abruptly increase in quantity after this point in a

statistically very significant manner, as shown by the results of a Wilcoxon Signed Rank test ($z = -2.949874$, $p = .003$).

	Before MLUw ~ 2.5	After MLUw ~ 2.5	%
Laura	1	20	4.8–95.2%
Gisela	0	9	0–100%
Martina	0	5	0–100%
Rosa	1	31	3.1–96.9%
Irene	0	85	0–100%
Koki	0	41	0–100 %
Kerstin	3	4	42.9–57.1%
Simone	2	7	22.2–77.8%
Josse	1	19	5–95%
Sarah	2	51	3.8–96.2%
Total	10	272	3.5–96.5%

Table 13: Production of Split CP-structures before and after MLUw ~ 2.5.

Note that the emphasis here is not on the exact MLUw ranges observed, but rather on the distinct *stages* observed. It is likely that this impressionistic MLUw ~2.5 threshold will not replicate itself in other children (note already that some variation is observed in the Dutch children, whose Stage 2 begins at ~3.5). The overall pattern would remain unaffected so long as Stage 3 remains a clearly separate stage.

Figure 5 plots the “explosive” emergence of Split CP-structures. The x-axis shows all files ordered by MLUw, and the y-axis the number of Split CP-structures produced. A change-point analysis was applied to this dataset. Change-point analysis is a statistical technique used to identify the point(s) in a time series at which a change occurs. It determines the number of significant changes observed in probability distributions and estimates the time of each change. All files from all languages were included, with Split CP-structure counts as the dependent variable. The red horizontal lines mark the estimated mean levels for each detected phase. The analysis reveals a first major change at approximately MLUw ~2.5 (file 170) – this is the same timeframe used for the comparison in **Table 13**.

The question then arises why, if some approaches posit cartographic representations throughout development, these do not materialise in the production data until significantly late stages, often *after* embedding structures emerge (**Table 11**). And conversely, CP-structures are nonetheless observable from the earliest word combinations. The abruptness and explosiveness

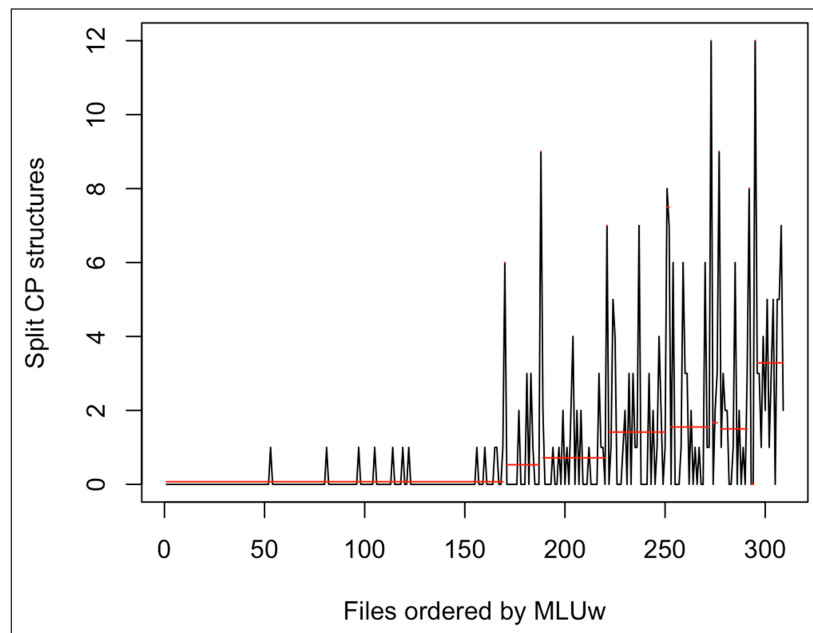


Figure 5: Change-point analysis of the distribution of Split CP-structures.

with which Split CP-structures emerge is also another important *explanandum* for any approach. This, then, raises an apparent representational paradox: how can we reconcile early coarse-grained CP-structures, with late Split CP-structures?

The finding from this third stage is striking in light of approaches that assume a UG-given spine to be accessible to the child (either partially at some stages, as in maturation, or totally, as in continuity). Based on the data presented, it appears that, while peripheral positions are used very early on, children operate initially on a coarse-grained discourse domain (interpreted as an emergent CP allowing a single head and specifier). Later developmental stages refine this coarse-grained CP and successively divide it into separate projections, such that this refined CP can now accommodate co-occurring CP-related structures. Crucially, a very similar generalisation was obtained in Soares (2006) for European Portuguese data, presenting a Derivational Complexity-based analysis (Jakubowicz 2011) whereby “the C category splits in the course of the derivation in function of its formal features, in order to create the necessary syntactic space enabling the verification of these features” (p. i).²¹

I therefore suggest the data is best explained if fine-grained syntactic heads of the sort in Rizzi (1997) are *emergent* and are not available as soon as CP becomes apparent at Stage 1, *contra* the expectations of maturational approaches (and recent cartographic continuity approaches like Westergaard 2009). Instead, they can be shown to systematically emerge at this second stage, as part of a refinement process over developmentally-earlier structures (a single-projection CP). The

²¹ Thanks very much to Carla Soares-Jesel for drawing my attention to this work.

data in this paper offers the first *preliminary* developmental evidence for this claim. Significantly, this generalisation also corroborates independent syntactic and biolinguistic work advocating for an ‘emergent cartography’; i.a., Ramchand & Svenonius (2014), Svenonius (2016), Scontras et al. (2017), Mišmaš et al. (2018), Leivada & Westergaard (2019), Marušič et al. (2019), Larson (2021) and Ramchand (2023).²² As a result, the following tentative empirical generalisation surfaces:

Generalisation 3 (Cartography is Emergent). Evidence for cartographic-type structure within CP systematically and abruptly emerges at a later developmental stage, elaborating on developmentally-prior structure (a “basic” CP).

A clear alternative hypothesis at this point, however, may be that the patterns observed are due to performance considerations. One notable possibility would be that Split CP-structures emerge late simply because they may be longer than basic CP-structures and thus might be expected to require higher MLUw. I give some brief results here to argue that relative length of the two diagnostics sets is likely not the driving factor of the patterns attested.

Via fixed-effects logistic regression, I addressed the possibility that the main predictor of presence/absence of Split CP utterances is their relative length, as opposed to Age and MLUw;²³ specifically, we ask if, having partialled out the effect of length in our model, Age and/or MLUw are still significant predictors. A sample of 264 utterances with CP-structures and with Split CP-structures from 5 children was extracted, including data for all relevant MLUw stages studied. The type of diagnostic reflected in an utterance (CP vs. Split CP) was coded as a binary dependent variable. The length (measured in words) of each utterance was gathered in a LENGTH variable, as well as the MLUw and AGE (in months) associated with each utterance.

Model comparisons with ANOVAs established that the best model fit included all three variables in the model as fixed effects (MLUw, AGE and LENGTH). Child was not included as a random effect; the AIC (Akaike information criterion) of the baseline model was lower than the model with random intercepts (322.4768 vs. 324.4768, respectively). The Model Likelihood Ratio Test also showed that the mixed-effects baseline model does not explain more variance ($p = 1$). As a result, the inclusion of random-effect structure in the model was not justified; I report here the results of the fixed-effects model. Thus, I fitted a logistic model (estimated using ML) to predict the use of Split CP utterances given these three variables (formula: `splitcp ~ mlu + length + age`). The effect of MLUw is *highly statistically significant* and positive ($\beta = 1.23$, 95% CI [0.63, 1.88], $p < .001$; Std. $\beta = 0.87$, 95% CI [0.44, 1.32]). Age is likewise *highly statistically*

²² See also Lee & Cournane (2019), Cournane & Klævik-Pettersen (2023) for other relevant work on diachronic syntax.

²³ Note that MLUw is crucially distinct from Length here: MLUw is a measure of how many words utterances have on average in a file, across *all utterances*. MLUw will increase as age increases, *irrespective* of whether the change is driven purely by sentence length or by grammatical changes. The alternative hypothesis at stake here is whether the difference between CP- and Split-CP-structures is *just* a difference in the length of these specific utterances (i.e., whether Length explains all the variance).

significant and positive ($\beta = 0.08$, 95% CI [0.03, 0.13], $p < .001$; Std. $\beta = 0.59$, 95% CI [0.25, 0.95]). Crucially, the effect of Length is *statistically non-significant* and positive ($\beta = 0.04$, 95% CI [-0.10, 0.18], $p = 0.563$; Std. $\beta = 0.10$, 95% CI [-0.24, 0.45]). This indicates both MLUw and Age are strong predictors of use/absence of Split CP-structures, but Length is *not*.

Therefore, I discard a length-based account of Generalisation 3, as it does not receive support from this data. Note also that an additional requirement on a performance-only account of these patterns is an explanation of the “explosiveness” facet of Split CP-structures. This observation aligns with the conclusions in Snyder (2007; 2021), who independently argues for the development of English verb-particle constructions that “explosive” changes in production (“decisive”, in his terms) are signs of a potential *grammatical* change. At any rate, however, establishing the role of other possible performance effects in the acquisition of Split CP-structures, above and beyond the length factor considered here, is a very important future task.

An anonymous reviewer also notes that the fact that many Split CP-structures are of the form Top > Wh (in Romance) (see p. 12), wherein both CP-constituents have been *moved*, may bear some significance under intervention-based accounts of syntactic development (e.g., Moscati 2023). Under these approaches, (at least some) Split CP-structures may be ruled out early on, not because of lack of CP-structure, but because Relativised Minimality (Rizzi 2004) reduces movement possibilities for children – as has been shown for the acquisition of object relatives and object wh-questions (Friedmann et al. 2009). This is a plausible objection to the account proposed here; however, it still overgenerates the kinds of Split CP-structures that should be produced earlier on. Specifically, it only predicts dislocations that generate a surface order deviating from Subject > Object to be problematic, i.e., $O_{[TOP]} S_{[WH/FOC]}$. This would then predict $S_{[TOP]} O_{[WH/FOC]}$ to be earlier-acquired.

The developmental patterns in Generalisation 3, however, suggest a much wider developmental ‘gap’, namely *complete* absence of Split CP-structures at earlier stages, either intervention-inducing or not. Indeed, based on a manual look at the Top > Wh structures collected, only 4 appear to involve true $O_{[TOP]} S_{[WH/FOC]} V$ orders, i.e., intervention configurations. Object topics but with non-subject wh-words (*where*, *how* and *why*) seem more common, in contrast (22 utterances attested). The same pattern is in fact reported for the acquisition of object topicalisation in German, only object topics that induce intervention effects are later-acquired (contrast section 3.4.2 with Ruigendijk & Friedmann 2017). This therefore corroborates that intervention *does* play a role in our patterns, as expected by Moscati (2023): Split CP-structures with intervention are barely attested. Even so, non-intervention configurations *are* produced abundantly (see examples in (52)–(54)), but nonetheless remarkably *late* – this latter pattern remains unexplained by syntactic intervention. I suggest, therefore, that intervention in the sense of Moscati (2023) only plays a *partial* role in the patterns at stake, in a way which is perfectly compatible with the broad proposals of this paper.

I now discuss the theoretical implications of these results in the next section and adopt a *representational difference* account of the contrast in Stage 1 vs. 2.

4 Discussion and implications

4.1 Theoretical implications: on the need for ontological flexibility

The goal of this paper was to identify potential ways in which incorporating granularity considerations in developmental theorising may be instructive. The results of the corpus study challenge the predictions of maturational and continuity approaches in various respects. Recall again the generalisations obtained, repeated here:

- i. **Generalisation 1** (Early Acquisition of CP). CP-structures emerge early on in the developmental data.
- ii. **Generalisation 2** (Structural Height and Acquisition Mismatch). There is a dissociation between structural height and order of emergence. Acquisition does not proceed successively upwards; some syntactically very high elements emerge early.
- iii. **Generalisation 3** (Cartography is Emergent). Evidence for cartographic-type structure within CP systematically and abruptly emerges at a later developmental stage, elaborating on developmentally-prior structure (a “basic” CP).

On the one hand, Generalisation 1 undermines the feasibility of bottom-up maturational approaches. The variety of early CP-structures observed suggests that a comprehensive model of these patterns should recognise a key insight of both continuity and inward-growing approaches: CP is an early development. At the same time, Generalisation 2 indicates that this early CP cannot be achieved by recapitulating a given cartographic spine in any directionality. This is at odds with recent approaches like the Growing Trees Hypothesis. The combination of Generalisations 1–2, then, underscores the fragility of a cartographic maturational approach based exclusively on *directionality*: irrespective of directionality assumed (bottom-up, inwardly), some structures are predicted to be late (e.g., topics/complementisers for a bottom-up approach), but turn out to be remarkably early developments in several children. Directionality of emergence is therefore not a productive tool with which to rationalise the empirical patterns here.

Stepping back, the developmental trends share many similarities with production and behavioural data reported for other languages supporting an early CP: e.g., for wh-questions (i.a., Santelmann 2003, for Swedish; Soares 2003, for European Portuguese; Saikkonen 2018, for Finnish; Seidl et al. 2003, and Perkins & Lidz 2021, for English), V2 (i.a., Boser et al. 1992, for German; van Kampen 2010, for Dutch; Westergaard 2009, for Norwegian), topicalisations and focalisations (Tsimpli 2005, for Greek; de Cat 2009, for French). If (some) interactional language is also hosted in the CP-domain (or a higher Speech Act domain), Shirai et al. (2000) and Liu &

Muramatsu (2025) also report very early emergence of (some) Mandarin and Japanese Sentence-Final particles; and Heim (2023) and Stoicescu (2025) provide data from English question tags and illocutionary complementisers in Romanian, respectively. This suggests that there is significant independent support for several of the claims adduced here.

The above leaves us with three extant approaches capable of capturing Generalisations 1–2: neo-emergentism, inward-growing approaches (e.g., Galasso 2003; Heim & Wiltschko 2025) and continuity. All expect early development of CP and do not make explicit commitments as to the order of emergence of CP-structures based on a cartographic left periphery.

However, Generalisation 3, I argue, requires, at least, a partial *departure* from all current approaches operating with innate categories, both maturation and continuity approaches. The empirical data presented forces us to assume some early representation of the CP-domain, but this representation cannot be finer-grained than Chomsky’s Core Functional Categories at this earlier stage, otherwise the theory overgenerates. Specifically, the generalisations force a theoretical approach which assumes some early knowledge of the CP-domain that does not develop into a cartographic-type left periphery until later. Development cannot simply recapitulate tree geometry for this effect to be generated. It would necessitate categorial systems and their granularity to be “malleable” during development, an analysis which is incompatible with proposed maturational approaches assuming exclusively innate categorial priors, such as a cartographic spine. The maturational proposals tested here, then, either undergenerate (assuming *no* CP-knowledge at the first stages of acquisition) or overgenerate (assuming *too fine-grained* knowledge of the CP-domain as soon as the latter emerges). I conclude, then, that a maturational approach is too rigid for this dataset.

Notice, additionally, that non-cartographic approaches do not technically face this contradiction, by virtue of assuming more impoverished clausal spines (such as Heim & Wiltschko’s 2025 Inward Growing Hypothesis and the earlier “Empty Middle” approach by Galasso 2003; or early versions of continuity). However, they arguably do not possess the tools to account for the contrast between Stage 1 and Stage 2. While not necessarily ruled out as possible developmental theories, further elaboration would be required to accommodate the patterns in Stage 3. Put differently, the three problems laid out in what follows would still apply to them. *Mutatis mutandis*, a similar point holds for continuity approaches: the abrupt increase in Split CP productions is unaddressed. This also carries over to other emergentist approaches that make developmental predictions based instead on “derivational ordering or timing”, such as Diercks & Bossi’s (2021) and Diercks et al.’s (2023) Developmental Minimalist Syntax. To be clear, the problem is *not* restricted to maturational approaches, but its paradoxical nature becomes most problematic in this type of theory.

With this in mind, I now propose to rethink these challenging empirical points by incorporating *granularity* into our analytical toolkit, in place of directionality, to address the

shortcomings in existing work; this addresses the shortcomings in existing work. Firstly, we can synthesise the ensuing discussion in terms of three interrelated (but dissociable) problems that have to be addressed, given this data:

i. **The Directionality Problem**

Developmental stages cannot straightforwardly be characterised as recapitulating a UG-given tree, either bottom-up or inwardly.

ii. **The Granularity Mismatch Problem**

Child grammars and adult grammars do not (always) operate at the same level of granularity. Acquisition may be characterised by *increasing granularity*, thus bridging the initial existing “gap” with the adult system.

iii. **The Comparability Problem**

The fine-tuning and structural homology patterns detected are not readily accounted for by innate categorial ontologies. Child categories may not always have direct counterparts in an adult-like UG-given spine.

I propose here that these problems are potentially serious for existent bottom-up maturational approaches, especially those of a cartographic nature. They can be resolved by dropping two assumptions in turn: firstly, foregoing an adult-like cartographic UG-given spine as the point of departure for the child and, by implication, a maturational mechanism that dictates the development of this latter spine (i.e., its *developmental directionality*). Doing so immediately lends us the additional ontological flexibility needed to account for (i)–(iii). In a neo-emergentist approach to acquisition, later developmental stages refine developmentally-prior categories (structural homology), leading to the *increasing granularity* patterns described. It also becomes misleading to describe development as recapitulating an innate spine in any given directionality and to assume that child categorial systems always have to be in direct correspondence with the categories postulated in adult-like UG-given spines (see again Heim & Wiltchko 2025: 17, for an approach with a similar ontological direction, applied to speech-act syntax: “these layers [early Speech Act layers – NBM] of structure are qualitatively different in child and adult language. [...] Specifically, we propose that the child gradually expands the complexity of the spine to include all the adult layers of structure”). In effect, I propose that only an approach that assumes categories to be *ontologically flexible* can readily capture the patterns in Generalisations 1–3.²⁴

²⁴ There still remains the possibility that an innate *template* of conceptual categories guides this emergence process (e.g., the *classification* → *point of view* → *anchoring* → *linking* template in Song 2019, from Wiltchko 2014, which mediates all granularity levels; see also Ramchand & Svenonius 2014). Additionally, continuity and inward-growing approaches could be made compatible with our data so long as emergent refinements on early-existing structures are theorised. I leave this open. The key argument here is that categorial divisions would have to be at least partly emergent to capture the patterns.

As a result, no such contradiction or problems arise with Biberauer & Roberts' (2015) successive-division learning path, which by design expects changes in representational granularity as development unfolds. The results of the corpus study are therefore consistent with it, as it correctly assumes that core macroparametric structural properties like a coarse-grained, "basic" representation of CP (e.g., Phasal or CFC) will emerge early. Only after children have mastered this option, will they acquire fine-grained ("micro-parametric") knowledge requiring cartographic-type projections. Crucially, I am making no specific claims regarding how and when children get the *entire* cartographic structure in Rizzi (1997) in languages where it is required (recall the discussion at the end of section 3.2.2). Indeed, the exact richness of CP is something neo-emergentism expects to be subject to crosslinguistic variation (Biberauer & Roberts 2015). The nature of the empirical data is far too coarse to disambiguate precisely *how* articulated the children's CP-representations are. The key here is that the 'tipping point' detected by Generalisation 3 points to a potential refinement in children's CP-structures, now allowing featurally finer-grained, minimally part-cartographic structures. This was completely absent in Stage 1, which appeared to allow only a more generic, 'single-slot' CP.

The foregoing is also an argument against *syntactic* (domain-specific) maturation: the data is firstly incompatible with it, but, equally importantly, maturation alone also cannot straightforwardly generate increasing granularity patterns. Reconsidering the value of syntactic maturation as a notion with explanatory weight in a Three Factors context is a conceptually appealing move (in the spirit of Chomsky 2005): it is in line with more recent attempts to minimise the role of UG and it aims to derive developmental patterns from more fundamental properties of complex systems and cognition, as opposed to a (stipulative)²⁵ biological imperative in UG. Specifically, levels of granularity are often suggested to be an important property of biological systems and cognition more broadly (see various perspectives in Zadeh 1997; Ehresmann & Vanbremeersch 2007; Chen & Du 2015; Lorkowski & Kreinovich 2015; Wang 2017; Wang et al. 2017; Kwisthout & van Rooij 2020). Smith et al. (2020) have also implemented computational simulations of structure learning based on the (Bayesian) active inference framework, showing that concept acquisition from coarse- to fine-grained may facilitate better results. In particular, Kwisthout (2013), Kwisthout et al. (2017), Rutar et al. (2022a; b) also critically discuss the need for state-space granularity in Bayesian models, and Ward et al. (2023) address its need in developmental frameworks. The importance attributed to 'detailedness' and 'granularity' considerations in biology and (computational) cognitive science thus suggests that granularity could be observable in the acquisition or shaping of language. There is some tentative evidence

²⁵ Maturation provides a possible description of some developmental patterns, but it is arguably not explanatory. Among other issues, it does not explain why functional spines should develop in the way they are proposed to (e.g., bottom-up, inwardly), nor does it address why the partitions of the spine into stages are the ones they are. It simply encodes the relevant developmental pathway by brute force (see Lust 1999: 124–128, in particular).

for this outside of syntax, see Mervis & Crisafi (1982), Xu & Tenenbaum (2007) and Murphy (2016) on lexical/concept acquisition; Krifka (2002; 2009), Cummins et al. (2012) and Enguehard (2018) on semantics/pragmatics; and Drescher (2009) and Nie et al. (2019) on phonology.

I reiterate again that I do not claim to have offered definite evidence for Generalisation 3 – this is necessarily a broader enterprise beyond the scope of this paper (see section 3.2). I do nonetheless treat the patterns emerging from the data discussed as unexpected by existing approaches and as worthy of explanation. The objectives of this section, then, are, firstly, to present a novel asymmetry and, secondly, to suggest a promising underexplored theoretical account of the empirical skew in the realisation of CP-structures. This spirit defines a new research agenda for the acquisition of CP.

In general, then, understanding development in this neo-emergentist way proves, at the very least, productive – it does not just *accommodate* the data observed, but it *predicts* all patterns: (i) early-established formal distinctions serving as the basis for later developmental stages, (ii) developments towards ever more language-specific categorial granularity (as in complex emergent systems more generally, Gibb et al. 2019) and (iii) the reflexes of these changes in the granularity of developing grammars manifesting abruptly and explosively in the production data.

4.2 Limitations and future extensions

The present work is nonetheless limited in several ways, given the number of children and languages studied. Nonetheless, it lays the ground for future work on the role of UG, maturation and granularity in development. Several tasks that emerge from this paper represent the topic of ongoing research, beyond those already noted earlier. For example, the sample of both children and languages must be expanded, including adding new diagnostics as required by the languages studied and an analysis of their adult input data. Comparative work could prove particularly illuminating: similar timelines of acquisition for “split” or “cartographic-type” structures are presented in Mitrofanova (2018) for Russian PPs, Sanfelici & Gallina (2022) for Italian PPs, De Lisser et al. (2017) for the Jamaican Creole Tense-Aspect-Mood domain, among others, and for finer-grained speech-act syntax (Heim & Wiltschko 2025). Additionally, MLUw has proven to be a fairly stable metric in the 10 children presented; however, its role and stability must also be examined further with bigger samples (see, e.g., Oosthuizen & Southwood 2009; Ezeizabarrena & García Fernández 2017, for critical discussion on MLUw). Importantly, another research strand involves incorporating granularity considerations in experimental studies.

Finally, a future research strand includes establishing parallels with diachronic trends. Recent work on feature bundling, whereby diachronically more stable stages involve maximally bundled heads (e.g., a C head with [TOPIC], [FOCUS], [FIN], etc.); Lee & Cournane 2019; Klævik-Pettersen 2019a; Klævik-Pettersen 2019b; Cournane & Klævik-Pettersen 2023), makes predictions that are compatible with neo-emergentism. Other diachronic proposals are suggestively similar to the ones

here, e.g. Sluckin's (2025: 41) analysis for the development of V2/V3 systems initially involves "a maximally underspecified C-head for which the full set of features are yet to be posited" before "children acquiring Kiezdeutsch and SG [Standard German – NBM] [...] decompose C/T into C and T and also separate the lower and higher specifiers of C/T into C1 and C2". Under the present approach, then, we likewise expect "increases" and "decreases" in the elaboration of CP to be useful tools for the explanation of diachronic stages (see also Madaro et al. 2025) – in ways that may also insightfully map onto developmental data. Studying the diachrony of functional structure will inform us about preferred categorial trends and about which approaches are most suited for generating the relevant patterns.

5 Conclusion

This paper has argued that neo-emergentism offers a new framework to study the nature of language acquisition, one where categorial granularity is considered an *object of study*, rather than a theoretical prior. In this paper, I have provided empirical evidence to show why such a shift in analytical focus is productive, and how it opens an unexplored research strand on possible changes in granularity in development.

I identified problems with exclusively directionality-driven and fixed-granularity approaches via a multilingual corpus study. Our initial results indicate that CP emerges early across all children and languages, but reveal a novel generalisation, worthy of further investigation: production evidence for cartographic-type organisation in the CP-domain systematically emerges at a later stage. Additionally, I established an important mismatch between structural height and timing of development; some left-peripherally very high elements emerge very early.

I then argued that maturational and continuity approaches cannot capture the three generalisations introduced, as they generate theoretically contradictory requirements. This is because of a (somewhat implicit) assumption in approaches of this sort – *fixed* granularity – which amounts to stating that, *without some further theoretical elaboration*, these frameworks, by design, cannot replicate "shifts" in mental granularity. I emphasised how neo-emergentism offers a novel perspective on the data: I proposed to resolve this apparent contradiction by eschewing bottom-up syntactic maturation and innate cartographic categories and, crucially, by incorporating *granularity* in our analytical and conceptual toolkit. Emergent categories, construed in the neo-emergentist approach proposed, allow the crosslinguistic developmental patterns to be rationalised, without unnecessary stipulations. We can, with inward-growing and continuity approaches, maintain that the C-domain emerges early, but *also* concede that it is not cartographic in its first instantiation, this being a subsequent development.

This work, although preliminary in scope, has potentially significant implications for how developmental stages can be conceptualised and raises challenging questions about categorial

ontology and flexibility. It also has implications for new ways of approaching developmental data, by introducing granularity as a testable aspect of development, probed here via Split CP-structures. Irrespective of the ultimate answers to these questions, a main goal of this paper was to bring these issues to the forefront, such that future research can work towards enhancing our understanding of the variety of grammatical factors shaping acquisition.

Data availability

The additional files for this article can be found here: <https://osf.io/utf3b>. This repository contains 3 Appendices:

- Appendix A: Full tables containing the structures produced by every child.
- Appendix B: Quantitative analysis of CP-structures.
- Appendix C: Fixed-effects model building process and data collected.

Funding information

This research was funded by St John's College (Hayes MPhil Scholarship) and the Cambridge Trust (Cambridge Opportunity Master's Studentship) from 2022–23 and, from 2023-present, by the Arts and Humanities Research Council (UKRI) and St John's College (Open-Oxford Cambridge AHRC – St John's Doctoral Training Partnership; project reference 2890509).

Acknowledgements

I owe an incredible intellectual debt to Theresa Biberauer, who supervised the MPhil thesis on which this paper is based. My sincerest thanks go to her for her continuous enthusiasm, for her clear-headedness when approaching linguistic questions, and for encouraging me to write this paper. Anything good that may stem from this paper is a testament to her dedication to her students. This work was also presented as joint papers with Theresa at BUCLD 48, IGG 49 and GALA 16; I thank the audiences at all of these conferences for their comments. I am also very thankful to Dora Alexopoulou, Ashton Brown, Cécile de Cat, Karen Lahousse, Barbara Lust, James Morley, Roman Feiman, Emanuela Sanfelici, Itamar Schatz, Julia Schwarz, Carla Soares-Jesel, William Snyder, Virginia Valian and Bert Vaux for comments and/or help on various versions of this project. Thanks also go to audiences at various seminar groups at Cambridge, KU Leuven, Padua and Frankfurt. Last but not least, I would like to thank three anonymous reviewers for their constructive comments that helped improve this paper. Any remaining errors are entirely my own.

Competing interests

The author has no competing interests to declare.

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